

PART ONE

Organizations, Management, and the Networked Enterprise

CHAPTER 1

Information Systems in Global
Business Today

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CHAPTER 3

Information Systems, Organizations,
and Strategy

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Information Systems

PART ONE introduces the major themes of this book, raising a series of important questions: What is an information system, and what are its management, organization, and technology dimensions? Why are information systems so essential in businesses today? Why are systems for collaboration and social business so important? How can information systems help businesses become more competitive? What broader ethical and social issues are raised by widespread use of information systems?

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Information Systems in Global Business Today

LEARNING OBJECTIVES

After reading this chapter, you will be able to answer the following questions:

- 1-1** How are information systems transforming business, and why are they so essential for running and managing a business today?
- 1-2** What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?
- 1-3** What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?
- 1-4** How will MIS help my career?

CHAPTER CASES

Smart Stores Reinvent the Retail Space
Will the Coronavirus Pandemic Make Working from Home the New Normal?
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Changes in the Financial Industry: Adyen and Fintech

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Instructional Video:
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MyLab MIS

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eText with Conceptual Animations

Smart Stores Reinvent the Retail Space

Although there has been an upsurge in online shopping, retail stores are not going away. Some traditional retailers are fighting back by using information technology to provide new ways to bring people into physical stores or enhance their in-store experiences, even with new requirements for social distancing.

Acrelec, a French digital signage company, is piloting technology to help retailers manage store curbside pickups by customers placing orders online. Customers can use a retailer's smartphone app to indicate they are coming to pick up an order, and the Acrelec system will estimate when a customer will arrive at a particular store. Object-recognition cameras identify exactly when a customer's car arrives and where it is parked. Acrelec is especially useful for big-box retailers, grocery stores, and home-improvement stores.

Shelves have become more than just a surface for storing and displaying objects. New systems for "smart" shelves use proximity sensors, 3D cameras, microphones, RFID readers, and weight sensors to enable interactions between shoppers in physical stores and the shelves they're standing in front of. These systems can create a highly personalized shopping experience that fundamentally improves the way shoppers move inside physical stores.

Brands and retailers such as Pepsi, Walmart, and Albertsons are starting to use Smart Shelf by AWM to replicate the benefits of the online experience in physical retail environments. Using super-wide-angle low-light HD cameras, retailers deploying Smart Shelf are able to view and track their products in real-time. The solution improves operational efficiencies by highlighting specific shelves that need product stocking and allows for real-time on-shelf marketing to consumers. When retailers connect Smart Shelf to their mobile apps, they can help shoppers locate products themselves through their smartphones and tablets.

AWM Frictionless is a walk-in, walk-out solution enabling customers to shop as normal and check out by simply exiting the store. The system uses digital shelving and object-recognition cameras to keep track of which customers leave with which items. When customers enter a store, they are required to have mobile device and facial recognition scans, which allow the system to charge their digital accounts when they leave with purchases and receive a receipt via email or text message.



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In March 2020 AWM opened a low-contact cashierless micromarket called QuickEats at a luxury apartment community owned by Greenwood & McKenzie in Santa Ana California. QuickEats utilizes AWM Frictionless and features grab-and-go products such as sodas, water, juice drinks, sandwiches, cheese plates, fruit, and household cleaning items.

AWM Smart Shelf is able to personalize shoppers' experiences when they are in stores based on the items they pick up, even if they don't purchase them. For example, if a customer picks up a box of cookies and then puts it back, the retailer can use the system to offer a discount on the shelf beneath that item the next time the shopper encounters it in the store. Cofounder Kurtis Van Horn believes that Smart Shelf can provide the same level of customization and personalization as found in online shopping to brick-and-mortar stores.

AWM also offers an anonymous consumer behavior tracking application that can direct customers to other parts of a store using digital signage, enabling up-to-the-minute advertising and pricing. AWM solutions can be implemented in a wide range of store sizes and formats, from micromarkets, to convenience stores, to larger-format retailers. Technology is redefining the role of the shelf in retail marketing.

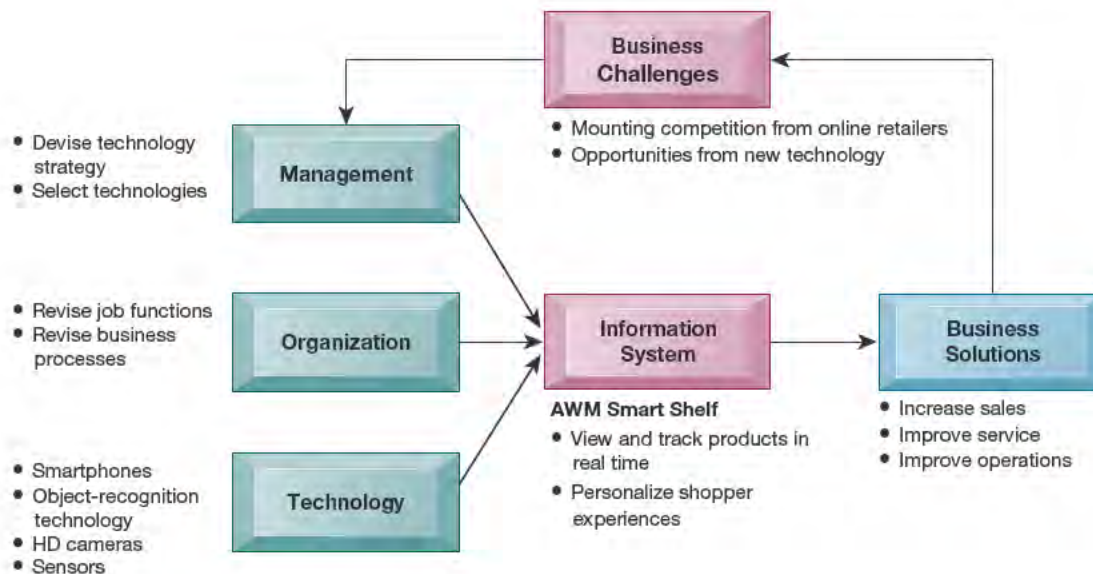
Sources: www.smartshelf.com, accessed April 29, 2020; AWM SmartShelf Launches Southern California's First Autonomous Micromarket in Santa Ana's Nineteen01 Community," *Businesswire*, March 24, 2020; Jared Council, "Retailers Hope In-Store Tech Will Keep Shoppers in Stores," *Wall Street Journal*, January 15, 2020.

The companies and technologies described here show how essential information systems are today. Today, retail stores are struggling to stay alive and relevant as more shoppers gravitate to online shopping and the Internet. One solution is to use leading-edge innovative information technology to provide new ways of drawing buyers into physical stores and making the in-store buying experience more efficient, safe, and pleasant. The information flows that drive these reimagined retail businesses have become much more digital, making use of mobile tools and object-recognition technology.

The chapter-opening diagram calls attention to important points raised by this case and this chapter. To compete more effectively against online retailers and take advantage of new technology solutions, brick-and-mortar retail stores are using innovative systems based on object-recognition technology, sensors, and smartphones. The use of leading-edge digital technologies to drive business operations and management decisions is a key topic today in the MIS world and will be discussed throughout this text.

It is also important to note that deploying information technology has changed the way customers of Acrelec and AWM Smart Shelf run their businesses. To effectively use new digital tools, these companies had to redesign jobs and procedures for gathering, inputting, and accessing information. These changes had to be carefully planned to make sure they enhanced efficiency, service, and profitability.

Here are some questions to think about: How do Acrelec's and AWM's systems change retail operations? How do they improve the customer experience?



1-1 How are information systems transforming business, and why are they so essential for running and managing a business today?

It's not business as usual in the global economy anymore. Information systems and technologies are transforming the global business environment. In 2019, global spending on information technology (IT) and IT services was nearly \$3.8 trillion (Gartner, 2019). In addition, firms spent another \$160 billion on management consulting and services—much of which involves redesigning firms' business operations to take advantage of these new technologies (Statista, 2020). In fact, most of the business value of IT investment derives from these organizational, management, and cultural changes inside firms (Saunders and Brynjolfsson, 2016). Figure 1.1 shows that between 1999 and 2019, capital investment in information technology consisting of IT equipment, software, and research and development (R&D) accounted for over 40 percent of US total capital spending. A similar pattern has occurred globally.

As managers, most of you will work for firms that are intensively using information systems and making large investments in information technology. You will certainly want to know how to invest this money wisely. If you make wise choices, your firm can outperform competitors. If you make poor choices, you will be wasting valuable capital. This book is dedicated to helping you make wise decisions about information technology and information systems.

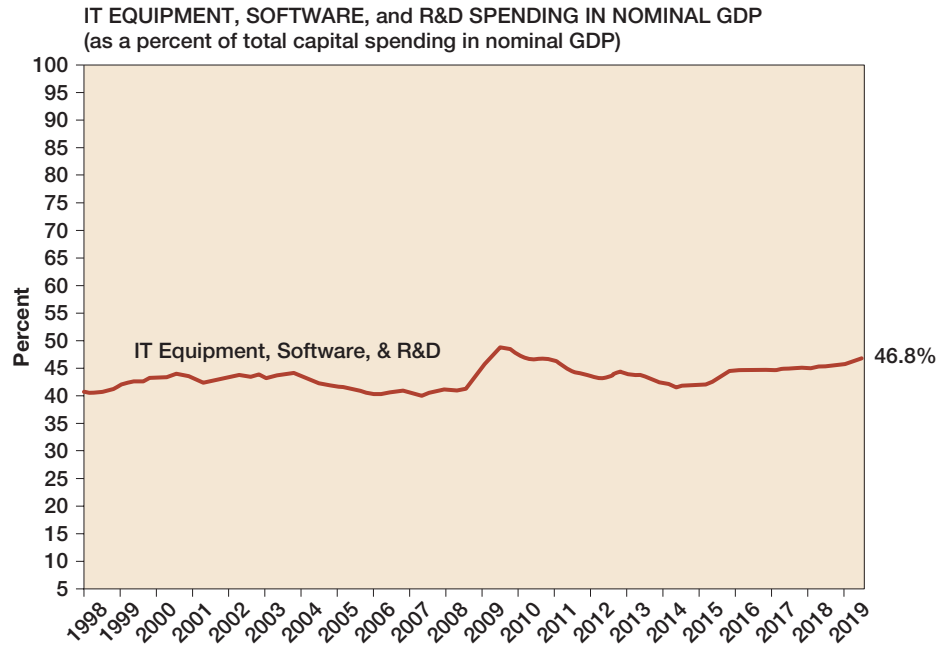
What's New in Management Information Systems?

Plenty. In fact, there's a whole new world of doing business using new technologies for managing and organizing. What makes the MIS field the most exciting area of study in schools of business is the continuous change in technology, management, and business processes. Five changes are of paramount importance.

FIGURE 1.1 INFORMATION TECHNOLOGY CAPITAL INVESTMENT

Information technology capital investment, defined as IT equipment, software and research and development (R&D) spending, amounted to over 40 percent of total US capital spending (in nominal GDP) between 1999 and 2019. A similar pattern has occurred globally.

Source: US Bureau of Economic Analysis.



IT Innovations A continuing stream of information technology innovations is transforming the traditional business world. Examples include the emergence of cloud computing, the growth of a mobile digital business platform based on smartphones and tablet computers, big data and the Internet of Things (IoT), business analytics, machine learning systems, and the use of social networks by managers to achieve business objectives. Most of these changes have occurred in the past few years. These innovations are enabling entrepreneurs and innovative traditional firms to create new products and services, develop new business models, and transform the day-to-day conduct of business. In the process, some old businesses, even industries, are being destroyed while new businesses are springing up.

New Business Models For instance, the emergence of online video services for streaming or downloading, such as Netflix, Apple TV Channels, and Amazon, has forever changed how premium video is distributed and even created. Netflix by early 2020 had attracted more than 167 million subscribers worldwide to what it calls the “Internet TV revolution.” Netflix has moved into premium TV show production with nearly 1,200 original shows in 2019, such as *American Vandal*, *Suburra*, *The Crown*, *Friends From College*, *House of Cards*, and *Orange Is the New Black*, challenging cable and broadcast producers of TV shows, and potentially disrupting cable network dominance of TV show production. Apple has struck deals with major Hollywood studios for recent movies and TV shows. A growing trickle of viewers are unplugging from cable and using only the Internet for entertainment.

E-commerce Expansion E-commerce sales worldwide amounted to nearly \$3.6 trillion in 2019 and are expected to approach \$5 trillion by 2021 (Lipsman, 2019). E-commerce is changing how firms design, produce, and deliver their products and services. E-commerce has reinvented itself again, disrupting the traditional marketing and advertising industry and putting major media and content firms

in jeopardy. Facebook and other social networking sites such as YouTube, Twitter, and Tumblr, along with Netflix, Apple Music, and many other media firms, exemplify the new face of e-commerce in the twenty-first century. They sell services. When we think of e-commerce, we tend to think of selling physical products. While this iconic vision of e-commerce is still very powerful and the fastest-growing form of retail worldwide, growing up alongside is a whole new value stream based on selling services, not goods. It's a services model of e-commerce. Growth in social commerce is spurred by powerful growth of the mobile platform: 98 percent of Facebook's users access the service from mobile phones and tablets. Information systems and technologies are the foundation of this new services-based e-commerce. Mobile retail e-commerce worldwide approached \$2.55 billion in 2020 (and is growing at almost 20 percent a year) (eMarketer, 2020).

Management Changes The management of business firms has changed: With new mobile smartphones, high-speed wireless Wi-Fi networks, and tablets, remote salespeople on the road are only seconds away from their managers' questions and oversight. Management is going mobile. Managers on the move are in direct, continuous contact with their employees. The growth of enterprise-wide information systems with extraordinarily rich data means that managers no longer operate in a fog of confusion but instead have online, nearly instant access to the really important information they need for accurate and timely decisions. In addition to their public uses on the web, social networking tools, wikis, and blogs are becoming important corporate tools for communication, collaboration, and information sharing.

Changes in Firms and Organizations Compared to industrial organizations of the previous century, new fast-growing twenty-first-century business firms put less emphasis on hierarchy and structure and more emphasis on employees taking on multiple roles and tasks and collaborating with others on a team. They put greater emphasis on competency and skills rather than position in the hierarchy. They emphasize higher-speed and more-accurate decision making based on data and analysis. They are more aware of changes in technology, consumer attitudes, and culture. They use social media to enter into conversations with consumers and demonstrate a greater willingness to listen to consumers, in part because they have no choice. They show better understanding of the importance of information technology in creating and managing business firms and other organizations. To the extent organizations and business firms demonstrate these characteristics, they are twenty-first-century digital firms.

You can see some of these trends at work in the Interactive Session on Organizations, which examines the impact on work and management as many companies had their employees work remotely during the coronavirus pandemic of 2020.

Globalization Challenges and Opportunities: A Flattened World

Prior to AD 1500, there was no truly global economic system of trade that connected all the continents on earth although there were active regional trade markets. After the sixteenth century, a global trading system began to emerge based on advances in navigation and ship technology. The world trade that ensued after these developments has brought the peoples and cultures of the world much closer together. The Industrial Revolution was really a worldwide phenomenon energized by expansion of trade among nations, making nations both competitors and collaborators in business. The Internet has greatly heightened the competitive tensions among nations as global trade expands

INTERACTIVE SESSION ORGANIZATIONS

Will the Coronavirus Pandemic Make Working from Home the New Normal?

As COVID-19 continued to spread around the globe, companies large and small started to make changes to the way they work, shuttering their offices and requiring most or all of their employees to work remotely from their homes.

- During the pandemic, ClearRisk, which offers integrated, cloud-based software solutions for claims, fleet, incident, and insurance certificate management had its entire staff working from home.
- Many large law firms, including Reed Smith, Baker McKenzie, and Nixon Peabody, closed offices and required work at home during the pandemic. The law firms emphasized that they could continue to serve clients despite office closings and remote work.
- OpenText Corp., a Canadian provider of enterprise information management products, plans to eliminate more than half of its 120 offices globally, with 2000 of its 15,000-person workforce working from home permanently.
- In mid-May 2020, Twitter Inc. notified employees that most of them could work from home indefinitely.

According to a recent MIT report, 34 percent of Americans who previously commuted to work stated that they were working from home by the first week of April 2020 due to the coronavirus outbreak. Prior to the pandemic, the number of people regularly working from home remained in the single digits, with only about 4 percent of the US workforce working from home at least half the time. However, the trend of working from home had been slowly gaining momentum thanks to advances in information technology for remote work and changes in corporate work culture. The coronavirus pandemic may mark a tipping point.

It's likely that many people who started working from home for the first time during the pandemic will continue to do so thereafter. New health guidelines about distancing will require some workplaces to expand to accommodate all their employees or to have a significant percentage of employees work permanently from home.

Information technologies driving these changes include broadband high-speed Internet connections,

laptop computers, tablets, smartphones, email, messaging, and videoconferencing tools. As companies shift their work from face-to-face to remote, video conferencing is becoming the new normal for meetings. People are trying to have good conversations, share critical information, generate new ideas, reach consensus, and make decisions quickly on this platform.

Although less than ideal for face-to-face interactions, videoconferencing is becoming more powerful and affordable. There are many options, including Skype, Skype for Business, Zoom, Microsoft Teams, Amazon Chime, BlueJeans, Cisco's WebEx, GoToMeeting, and Google Meet. Some business people are using the same tools they do in their personal communications, such as FaceTime and Facebook Messenger. (FaceTime now supports group video chat with up to 32 people.)

Video conference software such as WebEx and BlueJeans appears designed for more corporate uses. Other software such as Microsoft's Skype and Zoom feels more consumer-friendly and easier to set up, with free or low-cost versions suitable for smaller businesses. Skype works for video chats, calls, and instant messaging and can handle up to 50 people in a single video call. Skype allows calls to be recorded in case someone misses a meeting. Skype also provides file-sharing capabilities, caller ID, voicemail, a split view mode to keep conversations separate, and screen share on mobile devices.

Up to 1,000 users can participate in a single Zoom video call, and 49 videos can appear on the screen at once. Zoom includes collaboration tools like simultaneous screen-sharing and co-annotation, and the ability to record meetings and generate transcripts. Users can adjust meeting times, select multiple hosts, and communicate via chat if microphones and cameras are turned off.

There are definite benefits to remote work: lower overhead, more flexible schedules, reductions in employee commuting time and attrition rates, and increases in productivity. (Many companies reported that productivity did not suffer when employees worked at home during the pandemic.) According to Global Workplace Analytics, a typical company saves about \$11,000 per half-time telecommuter per year. Working remotely also poses challenges.

Not all employees have access to the Internet at home, and many work in industries that require on-site work. About 80 percent of American adults have high-speed broadband Internet service at home. However, according to a Pew Research Center study, racial minorities, older adults, rural residents, and people with lower levels of education and income are less likely to have in-home broadband service. In addition, one in five American adults access the Internet only through their smartphones. Employees with little children or small apartments find working at home more difficult.

Full-time employees are four times more likely to have remote work options than part-time employees. According to Global Workplace Analytics, a typical remote worker is college-educated, at least 45 years old, and earns an annual salary of \$58,000 while working for a company with more than 100 employees.

Although email and text messaging are very useful, they are not effective tools for communication compared to the information exchange and personal

connection of face-to-face conversations. Remote work also inhibits the creativity and innovative thinking that take place when people interact with each other face-to-face, and videoconferencing is only a partial solution. Studies have found that people working together in the same room tend to solve problems more quickly than remote collaborators, and that team cohesion suffers when members work remotely.

Sources: Dana Mattioli and Konrad Putzier, "The End of the Office," *Wall Street Journal*, May 16-17, 2020; Rani Molla, "This Is the End of the Office as We Know It," *Vox*, April 14, 2020; Josh Lowy, "Overcoming Remote Work Challenges," *MIT Sloan Management Review*, April 9, 2020; Cate Pye, "Coronavirus: What Does the 'New Normal' Mean for How We Work?" *Computer Weekly*, April 3, 2020; Lindsey Jacobson, "As Coronavirus Forces Millions to Work Remotely, the US Economy May Have Reached a 'Tipping Point' in Favor of Working from Home," *CNBC*, March 23, 2020; Derek Thompson, "The Coronavirus Is Creating a Huge, Stressful Experiment in Working from Home," *The Atlantic*, March 13, 2020; Kevin Roose, "Sorry, but Working from Home Is Overrated," *New York Times*, March 10, 2020.

CASE STUDY QUESTIONS

1. Define the problem described in this case. What are the management, organization, and technology issues raised by this problem?
2. Identify the information technologies used to provide a solution to this problem. Was this a successful solution? Why or why not?
3. Will working from home become the dominant way of working in the future? Why or why not?

and strengthened the benefits that flow from trade, and also created significant dislocations in labor markets.

In 2005, journalist Thomas Friedman wrote an influential book declaring the world was now flat, by which he meant that the Internet and global communications had greatly expanded the opportunities for people to communicate with one another and reduced the economic and cultural advantages of developed countries. The United States and European countries were in a fight for their economic lives, according to Friedman, competing for jobs, markets, resources, and even ideas with highly educated, motivated populations in low-wage areas in the less developed world (Friedman, 2007). This globalization presents you and your business with both challenges and opportunities.

A growing percentage of the economy of the United States and other advanced industrial countries in Europe and Asia depends on imports and exports. In 2019, an estimated 30 percent of the world economy resulted from foreign trade of goods and services, both imports and exports. Half of *Fortune* 500 US firms obtain nearly 50 percent of their revenue from foreign operations. For instance, more than 50 percent of Intel's revenues in 2019 came from overseas sales of its microprocessors.

It's not just goods that move across borders. So too do jobs, some of them high-level jobs that pay well and require a college degree. For instance, since 2000, the United States has lost an estimated 5 million manufacturing jobs to offshore, low-wage producers, so manufacturing is now a small part of US employment (less than 9 percent). In a normal year, about 300,000 service jobs move offshore to lower-wage countries, many of them in less-skilled information system occupations but also in tradable service jobs in architecture, financial services, customer call centers, consulting, engineering, and even radiology.

On the plus side, the US economy created 2.6 million new jobs in 2018. Employment in information systems and the other service occupations listed previously has rapidly expanded in sheer numbers, wages, productivity, and quality of work. Outsourcing has actually accelerated the development of new systems in the United States and worldwide by reducing the cost of building and maintaining them. In 2019 job openings in information systems and technologies far exceeded the supply of applicants.

The challenge for you as a business student is to develop high-level skills through education and on-the-job experience that cannot be outsourced. The challenge for your business is to avoid markets for goods and services that can be produced offshore much less expensively. The opportunities are equally immense. Throughout this book you will find examples of companies and individuals who either failed or succeeded in using information systems to adapt to this new global environment.

What does globalization have to do with management information systems? That's simple: everything. The emergence of the Internet into a full-blown international communications system has drastically reduced the costs of operating and transacting on a global scale. Communication between a factory floor in Shanghai and a distribution center in Rapid City, South Dakota, is now instant and virtually free. Customers can now shop in a worldwide marketplace, obtaining price and quality information reliably 24 hours a day. Firms producing goods and services on a global scale achieve extraordinary cost reductions by finding low-cost suppliers and managing production facilities in other countries. Internet service firms, such as Google and eBay, are able to replicate their business models and services in multiple countries without having to redesign their expensive fixed-cost information systems infrastructure. Briefly, information systems enable globalization.

The Emerging Digital Firm

All of the changes we have just described, coupled with equally significant organizational redesign, have created the conditions for a fully digital firm. A digital firm can be defined along several dimensions. A **digital firm** is one in which nearly all of the organization's *significant business relationships* with customers, suppliers, and employees are digitally enabled and mediated. *Core business processes* are accomplished through digital networks spanning the entire organization or linking multiple organizations.

Business processes refer to the set of logically related tasks and behaviors that organizations develop over time to produce specific business results and the unique manner in which these activities are organized and coordinated. Developing a new product, generating and fulfilling an order, creating a marketing plan, and hiring an employee are examples of business processes, and the ways organizations accomplish their business processes can be a source of competitive strength. (A detailed discussion of business processes can be found in Chapter 2.)

Key corporate assets—intellectual property, core competencies, and financial and human assets—are managed through digital means. In a digital firm, any piece of information required to support key business decisions is available at any time and anywhere in the firm.

Digital firms sense and respond to their environments far more rapidly than traditional firms, giving them more flexibility to survive in turbulent times. Digital firms offer extraordinary opportunities for more-flexible global organization and management. In digital firms, both time shifting and space shifting are the norm. *Time shifting* refers to business being conducted continuously, 24/7, rather than in narrow “work day” time bands of 9 a.m. to 5 p.m. *Space shifting* means that work takes place in a global workshop as well as within national boundaries. Work is accomplished physically wherever in the world it is best accomplished.

Many firms, such as Cisco Systems, 3M, and GE, are close to becoming digital firms, using the Internet to drive every aspect of their business. Most other companies are not fully digital, but they are moving toward close digital integration with suppliers, customers, and employees.

Strategic Business Objectives of Information Systems

What makes information systems so essential today? Why are businesses investing so much in information systems and technologies? In the United States, more than 25 million business and financial managers, and 36 million professional workers in the labor force rely on information systems to conduct business. Information systems are essential for conducting day-to-day business in most advanced countries as well as achieving strategic business objectives.

Entire sectors of the economy are nearly inconceivable without substantial investments in information systems. E-commerce firms such as Amazon, eBay, Google, and E*Trade simply would not exist. Today's service industries—finance, insurance, and real estate as well as personal services such as travel, medicine, and education—could not operate without information systems. Similarly, retail firms such as Walmart and Tesco and manufacturing firms such as General Motors and Siemens require information systems to survive and prosper. Just as offices, telephones, filing cabinets, and efficient tall buildings with elevators were once the foundations of business in the twentieth century, information technology is a foundation for business in the twenty-first century.

There is a growing interdependence between a firm's ability to use information technology and its ability to implement corporate strategies and achieve corporate goals (see Figure 1.2). What a business would like to do in five years often depends on what its systems will be able to do. Increasing market share, becoming the high-quality or low-cost producer, developing new products, and increasing employee productivity depend more and more on the kinds and quality of information systems in the organization. The more you understand about this relationship, the more valuable you will be as a manager.

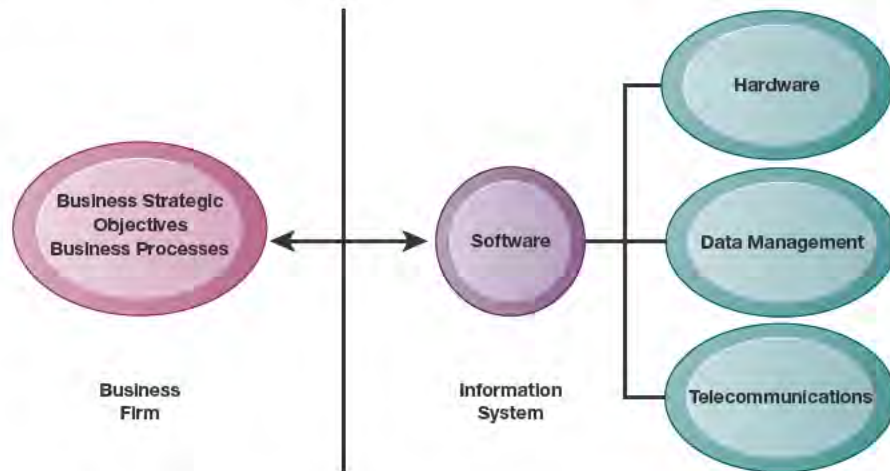
Specifically, business firms invest heavily in information systems to achieve six strategic business objectives: operational excellence; new products, services, and business models; customer and supplier intimacy; improved decision making; competitive advantage; and survival.

Operational Excellence

Businesses continuously seek to improve the efficiency of their operations in order to achieve higher profitability. Information systems and technologies are some of the most important tools available to managers for achieving higher

FIGURE 1.2 THE INTERDEPENDENCE BETWEEN ORGANIZATIONS AND INFORMATION SYSTEMS

In contemporary systems, there is a growing interdependence between a firm's information systems and its business capabilities. Changes in strategy, rules, and business processes increasingly require changes in hardware, software, databases, and telecommunications. Often, what the organization would like to do depends on what its systems will permit it to do.



levels of efficiency and productivity in business operations, especially when coupled with changes in business practices and management behavior.

Walmart, the largest retailer on earth, exemplifies the power of information systems coupled with state-of-the-art business practices and supportive management to achieve world-class operational efficiency. In fiscal year 2019, Walmart achieved \$524 billion in sales—nearly one-tenth of retail sales in the United States—in large part because of its Retail Link system, which digitally links its suppliers to every one of Walmart's stores. As soon as a customer purchases an item, the supplier monitoring the item knows to ship a replacement to the shelf. Walmart is the most efficient retail store in the industry.

New Products, Services, and Business Models

Information systems and technologies are a major enabling tool for firms to create new products and services as well as entirely new business models. A **business model** describes how a company produces, delivers, and sells a product or service to create wealth.

Today's music industry is vastly different from the industry a decade ago. Apple Inc. transformed an old business model of music distribution based on vinyl records, tapes, and CDs into an online, legal distribution model based on its own technology platform. Apple has prospered from a continuing stream of innovations, including the iTunes music service, the iPad, and the iPhone.

Customer and Supplier Intimacy

When a business really knows its customers and serves them well, the customers generally respond by returning and purchasing more. This raises revenues and profits. Likewise with suppliers—the more a business engages its suppliers, the better the suppliers can provide vital inputs. This lowers costs. How to really know your customers or suppliers is a central problem for businesses with millions of offline and online customers.

High-end hotels, such as the Mandarin Oriental hotel group, which operates hotels in Asia, Europe, and the Americas, exemplify the use of information

systems and technologies to achieve customer intimacy. These hotels use computers to keep track of guests' preferences, such as their preferred room temperature, check-in time, frequently dialed telephone numbers, and television programs, and store these data in a large data repository. Individual rooms in the hotels are networked to a central network server computer so that they can be remotely monitored and controlled. When a customer arrives at one of these hotels, the system automatically changes the room conditions, such as dimming the lights, setting the room temperature, or selecting appropriate music, based on the customer's digital profile. The hotels also analyze their customer data to identify their best customers and to develop individualized marketing campaigns based on customers' preferences.

Charles Tyrwhitt, a UK retailer specializing in dress shirts, and JCPenney, a US department store chain, exemplify the use of information systems to enable supplier and customer intimacy. Every time a dress shirt is bought one of their stores, the record of the sale appears immediately on computers in Hong Kong at their supplier. TAL Apparel, a contract manufacturer that produces one in six dress shirts sold in the United States. TAL runs the numbers through a computer model it developed and then decides how many replacement shirts to make and in what styles, colors, and sizes. TAL then sends the shirts to each store, bypassing the retailer's warehouses. These systems reduce inventory costs and ensure that what customers want are actually on the shelves.

Improved Decision Making

Many business managers operate in an information fog bank, never really having the right information at the right time to make an informed decision. Instead, managers rely on forecasts, best guesses, and luck. The result is over- or underproduction of goods and services, misallocation of resources, and poor response times. These poor outcomes raise costs and lose customers. In the past decade, information systems and technologies have made it possible for managers to use real-time data from the marketplace when making decisions.

For instance, Privi Organics Ltd., a leading Indian company that manufactures, supplies, and exports aroma chemical products worldwide, uses the Oracle Human Capital Management system for real-time insight into individual employee information—including performance rating and compensation history. The system helps managers make faster human resource decisions, such as promotions or transfers, by integrating all employee records across the organization. Managers are able to quickly review employee performance ratings for the previous three years and drill down into more details. A digital dashboard helps management view and monitor hiring status in multiple locations, such as the number of open positions and the time taken to fill these positions.

Competitive Advantage

When firms achieve one or more of these business objectives—operational excellence; new products, services, and business models; customer/supplier intimacy; and improved decision making—chances are they have already achieved a competitive advantage. Doing things better than your competitors, charging less for superior products, and responding to customers and suppliers in real time all add up to higher sales and higher profits that your competitors cannot match. Apple Inc., Walmart, and UPS, described later in this chapter, are industry leaders because they know how to use information systems for this purpose.

Survival

Business firms also invest in information systems and technologies because they are necessities of doing business. Sometimes these “necessities” are driven by industry-level changes. For instance, after Barclays Bank introduced the first automated teller machines (ATMs) in London in 1967, its competitors rushed to provide ATMs to their customers to keep up. Today, virtually all large banks around the world have regional ATMs and link to national and international ATM networks, such as Cirrus. Providing ATM services to retail banking customers is simply required in the retail banking business.

Most nations have statutes and regulations that create a legal duty for companies and their employees to retain records, including digital records. For instance, the European Council’s Regulation on Registration, Evaluation, Authorization, and Restriction of Chemicals (REACH) and the U.S. Toxic Substances Control Act (1976), which regulates the exposure of employees to toxic chemicals, require firms to retain records on employee exposure for a certain period of time. Many other pieces of national and regional legislation impose significant information retention and reporting requirements on US businesses. Firms turn to information systems to provide the capability to respond to these record management requirements.

1-2 What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?

So far we’ve used *information systems* and *technologies* informally without defining the terms. **Information technology (IT)** consists of all the hardware and software that a firm needs to use in order to achieve its business objectives. This includes not only computer machines, storage devices, and handheld mobile devices but also software, such as the Windows or Linux operating systems, the Microsoft Office desktop productivity suite, and the many thousands of computer programs that can be found in a typical large firm. “Information systems” are more complex and can be best understood by looking at them from both a technology and a business perspective.

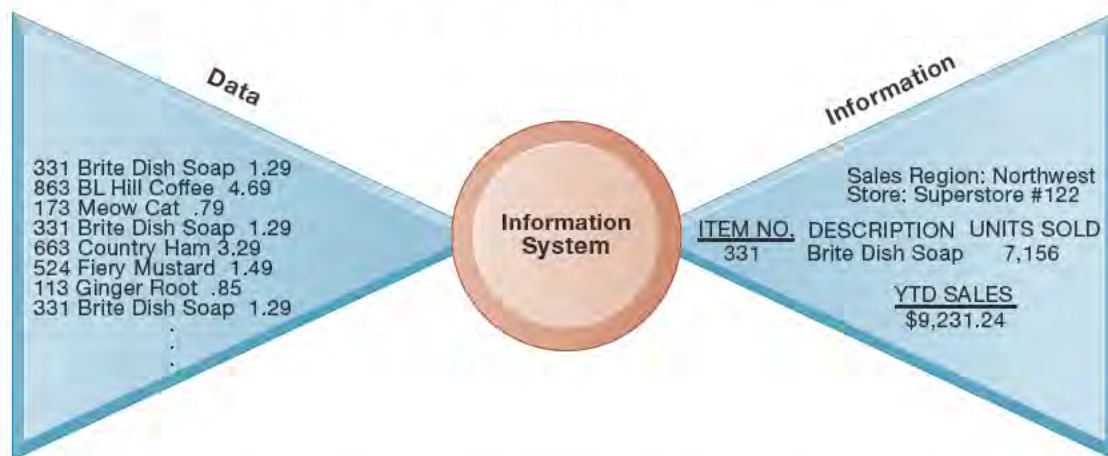
What Is an Information System?

An **information system** can be defined technically as a set of interrelated components that collect (or retrieve), process, store, and distribute information to support decision making and control in an organization. In addition to supporting decision making, coordination, and control, information systems may also help managers and workers analyze problems, visualize complex subjects, and create new products.

Information systems contain information about significant people, places, and things within the organization or in the environment surrounding it. By **information** we mean data that have been shaped into a form that is meaningful and useful to human beings. **Data**, in contrast, are streams of raw facts representing events occurring in organizations or the physical environment before they have been organized and arranged into a form that people can understand and use.

FIGURE 1.3 DATA AND INFORMATION

Raw data from a supermarket checkout counter can be processed and organized to produce meaningful information, such as the total unit sales of dish detergent or the total sales revenue from dish detergent for a specific store or sales territory.



A brief example contrasting information and data may prove useful. Supermarket checkout counters scan millions of pieces of data from bar codes, which describe each product. Such pieces of data can be totaled and analyzed to provide meaningful information, such as the total number of bottles of dish detergent sold at a particular store, which brands of dish detergent were selling the most rapidly at that store or sales territory, or the total amount spent on that brand of dish detergent at that store or sales region (see Figure 1.3).

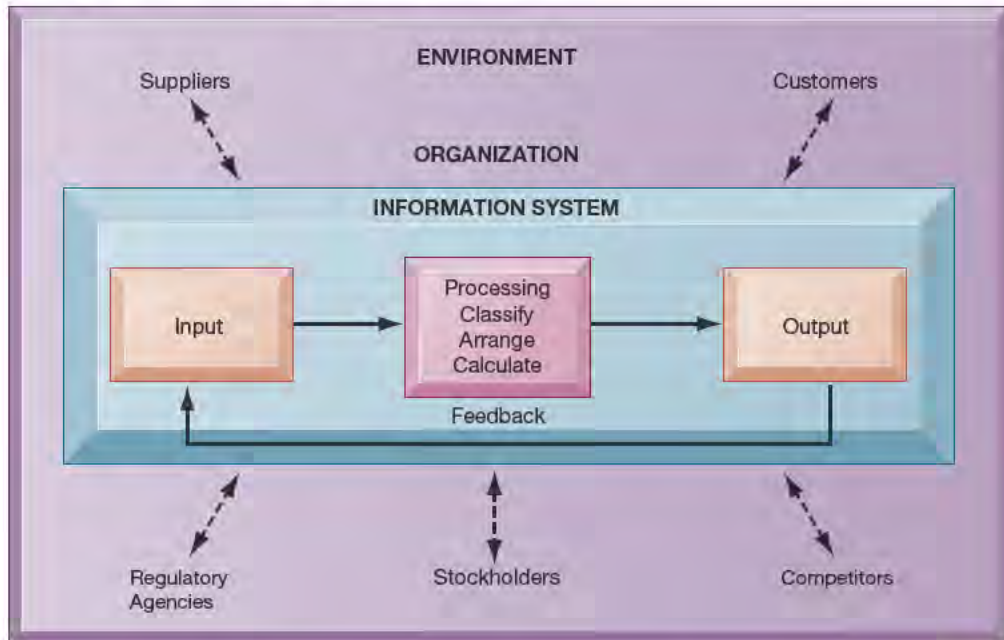
Three activities in an information system produce the information that organizations need to make decisions, control operations, analyze problems, and create new products or services. These activities are input, processing, and output (see Figure 1.4). **Input** captures or collects raw data from within the organization or from its external environment. **Processing** converts this raw input into a meaningful form. **Output** transfers the processed information to the people who will use it or to the activities for which it will be used. Information systems also require **feedback**, which is output that is returned to appropriate members of the organization to help them evaluate or correct the input stage.

In the AWM Smart Shelf system, input includes the digital results from scanning on-shelf products and store shelf identification codes, along with scanned images of items customers have selected for purchase. Computers store and process these data to keep track of the items on each shelf, the items each customer purchases or examines, and a history of the customer's purchases and items of interest. The system then determines what items on what shelves need restocking, and what items should be recommended to each customer. The system provides meaningful information such as all the items sold in a particular store or on a particular store shelf on a specific day, what items have been purchased by a specific customer, and which items need restocking.

Although computer-based information systems use computer technology to process raw data into meaningful information, there is a sharp distinction between a computer and a computer program on the one hand and an information system on the other. Computers and related software programs are the technical foundation, the tools and materials, of modern information systems. Computers provide the equipment for storing and processing information. Computer programs, or software, are sets of operating instructions that direct and control computer processing. Knowing how computers and computer

FIGURE 1.4 FUNCTIONS OF AN INFORMATION SYSTEM

An information system contains information about an organization and its surrounding environment. Three basic activities—input, processing, and output—produce the information organizations need. Feedback is output returned to appropriate people or activities in the organization to evaluate and refine the input. Environmental actors, such as customers, suppliers, competitors, stockholders, and regulatory agencies, interact with the organization and its information systems.



programs work is important in designing solutions to organizational problems, but computers are only part of an information system.

A house is an appropriate analogy. Houses are built with hammers, nails, and wood, but these do not make a house. The architecture, design, setting, landscaping, and all of the decisions that lead to the creation of these features are part of the house and are crucial for solving the problem of putting a roof over one's head. Computers and programs are the hammers, nails, and lumber of computer-based information systems, but alone they cannot produce the information a particular organization needs. To understand information systems, you must understand the problems they are designed to solve, their architectural and design elements, and the organizational processes that lead to the solutions.

Dimensions of Information Systems

To fully understand information systems, you must understand the broader organization, management, and information technology dimensions of systems (see Figure 1.5) and their power to provide solutions to challenges and problems in the business environment. We refer to this broader understanding of information systems, which encompasses an understanding of the management and organizational dimensions of systems as well as the technical dimensions of systems, as **information systems literacy**. **Computer literacy**, in contrast, focuses primarily on knowledge of information technology.

The field of **management information systems (MIS)** tries to achieve this broader information systems literacy. MIS deals with behavioral issues as well as technical issues surrounding the development, use, and impact of information systems used by managers and employees in the firm.

FIGURE 1.5 INFORMATION SYSTEMS ARE MORE THAN COMPUTERS

Using information systems effectively requires an understanding of the organization, management, and information technology shaping the systems. An information system creates value for the firm as an organizational and management solution to challenges posed by the environment.



Let's examine each of the dimensions of information systems—organizations, management, and information technology.

Organizations

Information systems are an integral part of organizations. Indeed, for some companies, such as credit reporting firms, there would be no business without an information system. The key elements of an organization are its people, structure, business processes, politics, and culture. We introduce these components of organizations here and describe them in greater detail in Chapters 2 and 3.

Organizations have a structure that is composed of different levels and specialties. Their structures reveal a clear-cut division of labor. Authority and responsibility in a business firm are organized as a hierarchy, or a pyramid structure. The upper levels of the hierarchy consist of managerial, professional, and technical employees, whereas the lower levels consist of operational personnel.

Senior management makes long-range strategic decisions about products and services as well as ensures financial performance of the firm. **Middle management** carries out the programs and plans of senior management, and **operational management** is responsible for monitoring the daily activities of the business. **Knowledge workers**, such as engineers, scientists, or architects, design products or services and create new knowledge for the firm, whereas **data workers**, such as secretaries or clerks, assist with scheduling and communications at all levels of the firm. **Production or service workers** actually produce the product and deliver the service (see Figure 1.6).

Experts are employed and trained for different business functions. The major **business functions**, or specialized tasks performed by business organizations, consist of sales and marketing, manufacturing and production, finance and accounting, and human resources (see Table 1.1). Chapter 2 provides more detail on these business functions and the ways in which they are supported by information systems.

An organization coordinates work through its hierarchy and through its *business processes*. Most organizations' business processes include formal rules that

TABLE 1.1 MAJOR BUSINESS FUNCTIONS

FUNCTION	PURPOSE
Sales and marketing	Selling the organization's products and services
Manufacturing and production	Producing and delivering products and services
Finance and accounting	Managing the organization's financial assets and maintaining the organization's financial records
Human resources	Attracting, developing, and maintaining the organization's labor force; maintaining employee records

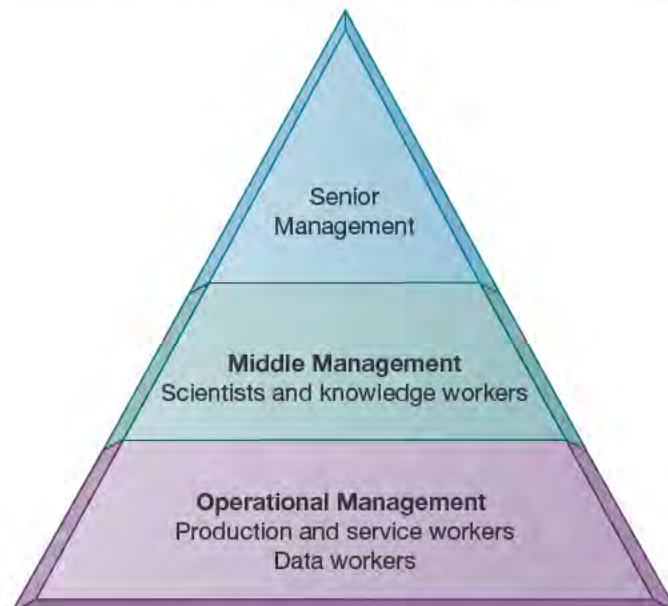
have been developed over a long time for accomplishing tasks. These rules guide employees in a variety of procedures, from writing an invoice to responding to customer complaints. Some of these business processes have been written down, but others are informal work practices, such as a requirement to return telephone calls from coworkers or customers, that are not formally documented. Information systems automate many business processes. For instance, how a customer receives credit or how a customer is billed is often determined by an information system that incorporates a set of formal business processes.

Each organization has a unique **culture**, or fundamental set of assumptions, values, and ways of doing things, that has been accepted by most of its members. You can see organizational culture at work by looking around your university or college. Some bedrock assumptions of university life are that professors know more than students, that the reason students attend college is to learn, and that classes follow a regular schedule.

Parts of an organization's culture can always be found embedded in its information systems. For instance, UPS's first priority is customer service, which is an aspect of its organizational culture that can be found in the company's package tracking systems, which we describe in this section.

FIGURE 1.6 LEVELS IN A FIRM

Business organizations are hierarchies consisting of three principal levels: senior management, middle management, and operational management. Information systems serve each of these levels. Scientists and knowledge workers often work with middle management.



Different levels and specialties in an organization create different interests and points of view. These views often conflict over how the company should be run and how resources and rewards should be distributed. Conflict is the basis for organizational politics. Information systems come out of this cauldron of differing perspectives, conflicts, compromises, and agreements that are a natural part of all organizations. In Chapter 3, we examine these features of organizations and their role in the development of information systems in greater detail.

Management

Management's job is to make sense out of the many situations faced by organizations, make decisions, and formulate action plans to solve organizational problems. Managers perceive business challenges in the environment, they set the organizational strategy for responding to those challenges, and they allocate the human and financial resources to coordinate the work and achieve success. Throughout, they must exercise responsible leadership. The business information systems described in this book reflect the hopes, dreams, and realities of real-world managers.

But managers must do more than manage what already exists. They must also create new products and services and even re-create the organization from time to time. A substantial part of management responsibility is creative work driven by new knowledge and information. Information technology can play a powerful role in helping managers design and deliver new products and services and redirecting and redesigning their organizations. Chapter 12 treats management decision making in detail.

Information Technology

Information technology is one of many tools managers use to cope with change.

Computer hardware is the physical equipment used for input, processing, and output activities in an information system. It consists of the following: computers of various sizes and shapes (including mobile handheld devices); various input, output, and storage devices; and telecommunications devices that link computers together.

Computer software consists of the detailed, preprogrammed instructions that control and coordinate the computer hardware components in an information system. Chapter 5 describes the contemporary software and hardware platforms used by firms today in greater detail.

Data management technology consists of the software governing the organization of data on physical storage media. More detail on data organization and access methods can be found in Chapter 6.

Networking and telecommunications technology, consisting of both physical devices and software, links the various pieces of hardware and transfers data from one physical location to another. Computers and communications equipment can be connected in networks for sharing voice, data, images, sound, and video. A **network** links two or more computers to share data or resources, such as a printer.

The world's largest and most widely used network is the **Internet**. The Internet is a global "network of networks" that uses universal standards (described in Chapter 7) to connect millions of networks in more than 230 countries around the world.

The Internet has created a new "universal" technology platform on which to build new products, services, strategies, and business models. This same technology platform has internal uses, providing the connectivity to link different systems and networks within the firm. Internal corporate networks based on Internet

technology are called **intranets**. Private intranets extended to authorized users outside the organization are called **extranets**, and firms use such networks to coordinate their activities with other firms for making purchases, collaborating on design, and other interorganizational work. For most business firms today, using Internet technology is both a business necessity and a competitive advantage.

The **World Wide Web** is a service provided by the Internet that uses universally accepted standards for storing, retrieving, formatting, and displaying information in a page format on the Internet. Web pages contain text, graphics, animations, sound, and video and are linked to other web pages. By clicking on highlighted words or buttons on a web page, you can link to related pages to find additional information and links to other locations on the web. The web can serve as the foundation for new kinds of information systems such as UPS's web-based package tracking system described below.

All of these technologies, along with the people required to run and manage them, represent resources that can be shared throughout the organization and constitute the firm's **information technology (IT) infrastructure**. The IT infrastructure provides the foundation, or *platform*, on which the firm can build its specific information systems. Each organization must carefully design and manage its IT infrastructure so that it has the set of technology services it needs for the work it wants to accomplish with information systems. Chapters 5 through 8 of this book examine each major technology component of information technology infrastructure and show how they all work together to create the technology platform for the organization. The Interactive Session on Technology describes some examples of the technologies used in computer-based information systems today.

UPS's global delivery system provides another example. UPS invests heavily in information systems technology to make its business more efficient and customer oriented. It uses an array of information technologies, including bar code scanning systems, wireless networks, large mainframe computers, handheld computers, the Internet, and many different pieces of software for tracking packages, calculating fees, maintaining customer accounts, and managing logistics.

Let's identify the organization, management, and technology elements in the UPS package tracking system. The organization element anchors the package tracking system in UPS's sales and production functions (the main product of UPS is a service—package delivery). It specifies the required procedures for identifying packages with both sender and recipient information, taking inventory, tracking the packages en route, and providing package status reports for UPS customers and customer service representatives.

The system must also provide information to satisfy the needs of managers and workers. UPS drivers need to be trained in both package pickup and delivery procedures and in how to use the package tracking system so that they can work efficiently and effectively. UPS customers may need some training to use UPS in-house package tracking software or the UPS website.

UPS's management is responsible for monitoring service levels and costs and for promoting the company's strategy of combining low cost and superior service. Management decided to use computer systems to increase the ease of sending a package using UPS and of checking its delivery status, thereby reducing delivery costs and increasing sales revenues.

The technology supporting this system consists of handheld computers, bar code scanners, desktop computers, wired and wireless communications networks, UPS's data center, storage technology for the package delivery data, UPS in-house package tracking software, and software to access the web. The result is an information system solution to the business challenge of providing a high level of service with low prices in the face of mounting competition.

INTERACTIVE SESSION TECHNOLOGY

Digital Transformation of Healthcare at Singapore's Jurong Health Services

Jurong Health Services (JurongHealth) is one of Singapore's six public healthcare clusters. Healthcare clusters provide holistic and integrated care when patients move from one care setting, like a clinic, to another, like a hospital. JurongHealth primarily manages the 700-bed Ng Teng Fong General Hospital, the 400-bed Jurong Community Hospital, and the Jurong Medical Center, all of which are located in western Singapore.

JurongHealth's goal is to provide transformative medical care for its patients through the use of innovative information technologies. Underscoring this commitment, in 2016 JurongHealth's Ng Teng Fong General Hospital became the first hospital in Singapore and the ASEAN region, and fifth in Asia-Pacific, to receive the highest level of the Healthcare Information and Management Systems Society (HIMSS) Electronic Medical Record Adoption Model Award, which measures a hospital's implementation of IT systems.

JurongHealth has integrated more than 50 healthcare IT systems as part of the Project OneCare initiative. The systems' implementation and integration took four years and has enabled the hospital to become paperless, chartless, and filmless. For instance, self-service kiosks now enable patients to register themselves merely by scanning their national identification cards and obtaining a queue number generated by the Enterprise Queue Management System. This unique number is used throughout the patient's visit for all service itineraries in the hospital. Patients refer to live screens located in the waiting areas that display a real-time queue status showing their turn. This system has not only helped JurongHealth to cut down on expenses but also to improve efficiency, as patients do not need different numbers for different services. It reduces waiting time and increases patient satisfaction.

Similarly, the Visitor Management System self-service kiosks enable visitors to scan their identification cards and register themselves to gain access to hospital wards. Visitors can also register themselves and obtain an e-pass from the Visitor Registration counters. The identification card or e-pass is scanned at the gantry when entering and leaving the ward. The gantry not only logs visitor information but also tracks staff, who must also use the same gantries to enter the wards. The Visitor Management

System enables the hospital to both control access to the wards and track or contact visitors and staff if necessary.

Another IT system implemented is the Warehouse Management System, which eliminates the tedious process of manually counting inventory. The system uses passive radio frequency identification (RFID) technology and a two-bin shelving system to automate inventory top-up requests and improve inventory management. Once the primary compartment of the storage bin is empty, the clinical staff transfers the relevant RFID tag into a drop-box, where the reader automatically sends a request for drug replenishment, thus avoiding stock-outs.

JurongHealth has also implemented a Real-Time Location Tracking System to automatically track patients and medical equipment using Wi-Fi triangulation, low frequency excitors, and about 6,000 active RFID tags attached to patients or medical equipment. These tags continuously communicate with the low frequency excitors to transmit data to the backend system for processing, allowing hospital staff to precisely locate patients and equipment, thus eliminating the need for tedious manual searching.

JurongHealth made a conscious effort to ensure that the different IT systems would not be stand-alone. The hospital thus implemented an integrated Electronic Medical Record (EMR) system that combines all the functional modules of the hospital in addition to being interfaced with 140 medical devices and equipment. Using the vendor-neutral Medical Devices Middleware Integration System, data from these medical devices is directly uploaded into the EMR system, so that the clinical staff no longer has to manually enter such readings, reducing the likelihood of charting errors. Being vendor-neutral also means the freedom to adopt best-of-breed individual modules as well as a lack of reliance on a single vendor.

Since the implementation of the EMR system, JurongHealth has continued to move forward, particularly in response to the COVID-19 pandemic. In September 2020, the hospital introduced a series of measures to make the facilities more secure and infection-free. For instance, they deployed a security robot to patrol the grounds and a second "Kenobi" (a security and concierge robot) to carry out thermal scanning of visitors. Drones have also been deployed

to carry out inspections and surveillance. In addition, as an immediate response to the pandemic, and as a longer-term solution to infection control, new touch-free technology has been introduced to operate patient and visitor lifts.

IT has played a key role in enabling JurongHealth to achieve its mission of providing world-class medical care at an affordable cost. As a result, JurongHealth has developed a reputation as a leading technology-driven healthcare provider as well as a role model not only in Singapore but also the entire region. This is also evident from its many accolades, which include awards for IT-driven transformation and for overall organizational transformation.

CASE STUDY QUESTIONS

1. What technologies are used by JurongHealth? What purpose do they serve?
2. Search the web for RFID. Suggest an example of how RFID can be used for locating and tracking people.

*Case contributed by Neerja Sethi and Vijay Sethi,
Nanyang Technological University*

Sources: "Combating Covid-19: A Birth of Innovations," The Straits Times, September 29, 2020; Salma Khalik, "Singapore on Track to Have One of World's Most IT-enabled Healthcare Systems," The Straits Times, June 2, 2017; JurongHealth, "Awards & Accolades," www.juronghealth.com.sg, accessed January 5, 2017; P. Bhunia, "The JurongHealth IT Journey—Integrating IT from the Ground-Up into a New Digital Hospital," opengovasia.com, November 13, 2016; IHIS, "Ng Teng Fong General Hospital Becomes First in Singapore and ASEAN to Achieve HIMSS Analytics EMRAM Stage 7 Award," Press Release, www.ihis.com, October 7, 2016; "Our Milestones," www.juronghealth.com.sg, October 2016; Ai Lei Tao, "Singapore's Jurong Health Services: Transforming Healthcare Through Data Technologies," computerweekly.com, April 16, 2016; A. Shukla, "Singapore Hospitals Deliver Enhanced Care to Patients by Integrating Their IT Systems," www.cio-asia.com, March 17, 2016; IHIS, "Integrated Healthcare IT Systems at Ng Teng Fong General Hospital and Jurong Community Hospital Win the Project of the Year Award at the SPMI Symposium 2016," Press Release, www.ihis.com, March 14, 2016; J. Kelleher, "JurongHealth Services CIO Discusses the Fully Integrated EMR Suite, Hospital ICT Systems and Achieving HIMSS EMRAM Stage 6," opengovasia.com, February 18, 2016.

3. What information systems are implemented by JurongHealth? Describe the input, processing, and output of any one such system.
4. Why are information systems important for JurongHealth?

It Isn't Just Technology: A Business Perspective on Information Systems

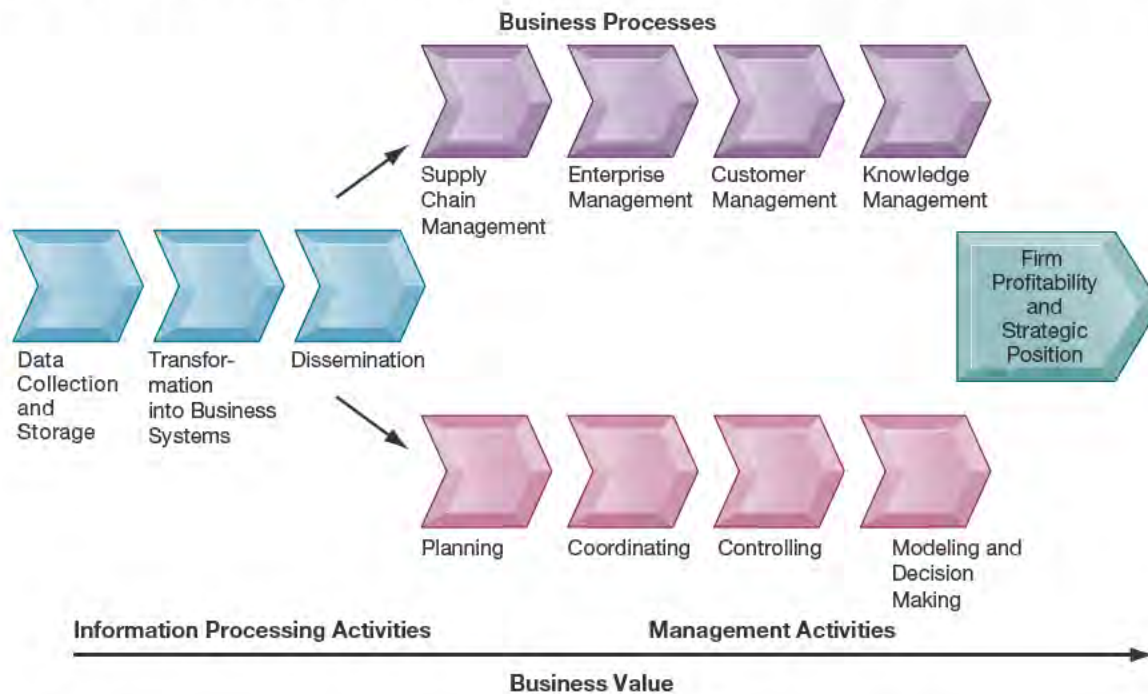
Managers and business firms invest in information technology and systems because they provide real economic value to the business. The decision to build or maintain an information system assumes that the returns on this investment will be superior to other investments in buildings, machines, or other assets. These superior returns will be expressed as increases in productivity, as increases in revenues (which will increase the firm's stock market value), or perhaps as superior long-term strategic positioning of the firm in certain markets (which will produce superior revenues in the future).

We can see that from a business perspective, an information system is an important instrument for creating value for the firm. Information systems enable the firm to increase its revenue or decrease its costs by providing information that helps managers make better decisions or that improves the execution of business processes. For example, the information system for analyzing supermarket checkout data illustrated in Figure 1.3 can increase firm profitability by helping managers make better decisions as to which products to stock and promote in retail supermarkets.

Every business has an information value chain, illustrated in Figure 1.7, in which raw information is systematically acquired and then transformed through various stages that add value to that information. The value of an information system to a business, as well as the decision to invest in any new information system, is, in large part, determined by the extent to which the system will lead to better management decisions, more efficient business processes, and higher

FIGURE 1.7 THE BUSINESS INFORMATION VALUE CHAIN

From a business perspective, information systems are part of a series of value-adding activities for acquiring, transforming, and distributing information that managers can use to improve decision making, enhance organizational performance, and, ultimately, increase firm profitability.



firm profitability. Although there are other reasons why systems are built, their primary purpose is to contribute to corporate value.

The business perspective calls attention to the organizational and managerial nature of information systems. An information system represents an organizational and management solution, based on information technology, to a challenge or problem posed by the environment. Every chapter in this book begins with a short case study that illustrates this concept. A diagram at the beginning of each chapter illustrates the relationship between a business challenge and resulting management and organizational decisions to use IT as a solution to challenges generated by the business environment. You can use this diagram as a starting point for analyzing any information system or information system problem you encounter.

Review the diagram at the beginning of this chapter. The diagram shows how the AWM Smart Shelf system helps solve the business problem of brick-and-mortar stores losing market share to online retailers. This system provides a solution that takes advantage of opportunities provided by new object-recognition and wireless digital technology.

AWM Smart Shelf digitally enables key business processes for inventory management, sales, and marketing, helping retailers improve their overall business performance. The diagram also illustrates how management, technology, and organizational elements work together to create the systems.

Complementary Assets: Organizational Capital and the Right Business Model

Awareness of the organizational and managerial dimensions of information systems can help us understand why some firms achieve better results from their information systems than others. Studies of returns from information

TABLE 1.2 COMPLEMENTARY SOCIAL, MANAGERIAL, AND ORGANIZATIONAL ASSETS REQUIRED TO OPTIMIZE RETURNS FROM INFORMATION TECHNOLOGY INVESTMENTS

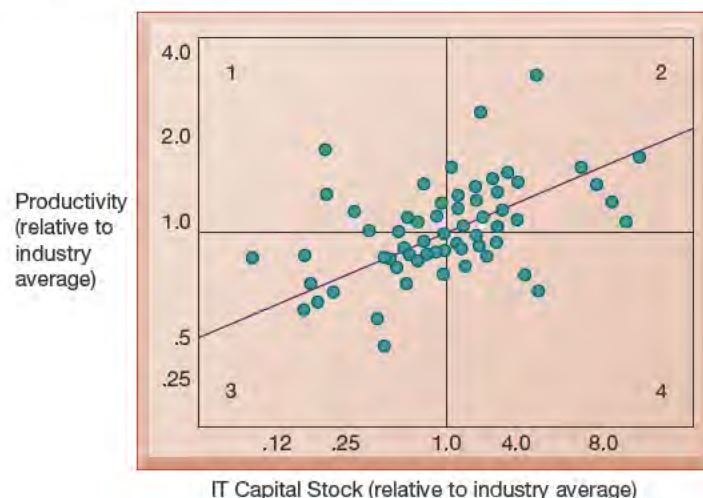
Organizational assets	Supportive organizational culture that values efficiency and effectiveness Appropriate business model Efficient business processes Decentralized authority Distributed decision-making rights Strong IS development team
Managerial assets	Strong senior management support for technology investment and change Incentives for management innovation Teamwork and collaborative work environments Training programs to enhance management decision skills Management culture that values flexibility and knowledge-based decision making
Social assets	The Internet and telecommunications infrastructure IT-enriched educational programs raising labor force computer literacy Standards (both government and private sector) Laws and regulations creating fair, stable market environments Technology and service firms in adjacent markets to assist implementation

technology investments show that there is considerable variation in the returns firms receive (see Figure 1.8). Some firms invest a great deal and receive a great deal (quadrant 2); others invest an equal amount and receive few returns (quadrant 4). Still other firms invest little and receive much (quadrant 1), whereas

FIGURE 1.8 VARIATION IN RETURNS ON INFORMATION TECHNOLOGY INVESTMENT

Although, on average, investments in information technology produce returns far above those returned by other investments, there is considerable variation across firms.

Source: Brynjolfsson, Erik, and Lorin M. Hitt. "Beyond Computation: Information Technology, Organizational Transformation, and Business Performance." *Journal of Economic Perspectives* 14, No. 4 (2000).



others invest little and receive little (quadrant 3). This suggests that investing in information technology does not by itself guarantee good returns. What accounts for this variation among firms?

The answer lies in the concept of complementary assets. Information technology investments alone cannot make organizations and managers more effective unless they are accompanied by supportive values, structures, and behavior patterns in the organization and other complementary assets. Business firms need to change how they do business before they can really reap the advantages of new information technologies.

Complementary assets are those assets required to derive value from a primary investment (Teece, 1998). For instance, to realize value from automobiles requires substantial complementary investments in highways, roads, gasoline stations, repair facilities, and a legal regulatory structure to set standards and control drivers.

Research indicates that firms that support their technology investments with investments in complementary assets, such as new business models, new business processes, management behavior, organizational culture, or training, receive superior returns, whereas those firms failing to make these complementary investments receive less or no returns on their information technology investments (Brynjolfsson, 2005; Brynjolfsson and Hitt, 2000; Laudon, 1974). These investments in organization and management are also known as **organizational and management capital**.

Table 1.2 lists the major complementary investments that firms need to make to realize value from their information technology investments. Some of this investment involves tangible assets, such as buildings, machinery, and tools. However, the value of investments in information technology depends to a large extent on complementary investments in management and organization.

Key organizational complementary investments are a supportive business culture that values efficiency and effectiveness, an appropriate business model, efficient business processes, decentralization of authority, highly distributed decision rights, and a strong information system (IS) development team.

Important managerial complementary assets are strong senior management support for change, incentive systems that monitor and reward individual innovation, an emphasis on teamwork and collaboration, training programs, and a management culture that values flexibility and knowledge.

Important social investments (not made by the firm but by the society at large, other firms, governments, and other key market actors) are the Internet and the supporting Internet culture, educational systems, network and computing standards, regulations and laws, and the presence of technology and service firms.

Throughout the book, we emphasize a framework of analysis that considers technology, management, and organizational assets and their interactions. Perhaps the single most important theme in the book, reflected in case studies and exercises, is that managers need to consider the broader organization and management dimensions of information systems to understand current problems as well as to derive substantial above-average returns from their information technology investments. As you will see throughout the text, firms that can address these related dimensions of the IT investment are, on average, richly rewarded.

1-3 What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

The study of information systems is a multidisciplinary field. No single theory or perspective dominates. Figure 1.9 illustrates the major disciplines that contribute problems, issues, and solutions in the study of information systems. In general, the field can be divided into technical and behavioral approaches. Information systems are sociotechnical systems. Though they are composed of machines, devices, and “hard” physical technology, they require substantial social, organizational, and intellectual investments to make them work properly.

Technical Approach

The technical approach to information systems emphasizes mathematically based models to study information systems as well as the physical technology and formal capabilities of these systems. The disciplines that contribute to the technical approach are computer science, management science, and operations research.

Computer science is concerned with establishing theories of computability, methods of computation, and methods of efficient data storage and access. Management science emphasizes the development of models for decision-making and management practices. Operations research focuses on mathematical techniques for optimizing selected parameters of organizations, such as transportation, inventory control, and transaction costs.

Behavioral Approach

An important part of the information systems field is concerned with behavioral issues that arise in the development and long-term maintenance of information systems. Issues such as strategic business integration, design, implementation,

FIGURE 1.9 CONTEMPORARY APPROACHES TO INFORMATION SYSTEMS

The study of information systems deals with issues and insights contributed from technical and behavioral disciplines.



utilization, and management cannot be explored usefully with the models used in the technical approach. Other behavioral disciplines contribute important concepts and methods.

For instance, sociologists study information systems with an eye toward how groups and organizations shape the development of systems and also how systems affect individuals, groups, and organizations. Psychologists study information systems with an interest in how human decision makers perceive and use formal information. Economists study information systems with an interest in understanding the production of digital goods, the dynamics of digital markets, and how new information systems change the control and cost structures within the firm.

The behavioral approach does not ignore technology. Indeed, information systems technology is often the stimulus for a behavioral problem or issue. But the focus of this approach is generally not on technical solutions. Instead, it concentrates on changes in attitudes, management and organizational policy, and behavior.

Approach of This Text: Sociotechnical Systems

Throughout this book, you will find a rich story with four main actors: suppliers of hardware and software (the technologists); business firms making investments and seeking to obtain value from the technology; managers and employees seeking to achieve business value (and other goals); and the contemporary legal, social, and cultural context (the firm's environment). Together these actors produce what we call *management information systems*.

The study of management information systems (MIS) arose to focus on the use of computer-based information systems in business firms and government agencies. MIS combines the work of computer science, management science, and operations research with a practical orientation toward developing system solutions to real-world problems and managing information technology resources. It is also concerned with behavioral issues surrounding the development, use, and impact of information systems, which are typically discussed in the fields of sociology, economics, and psychology.

Our experience as academics and practitioners leads us to believe that no single approach effectively captures the reality of information systems. The successes and failures of information systems are rarely all technical or all behavioral. Our best advice to students is to understand the perspectives of many disciplines. Indeed, the challenge and excitement of the information systems field are that it requires an appreciation and tolerance of many different approaches.

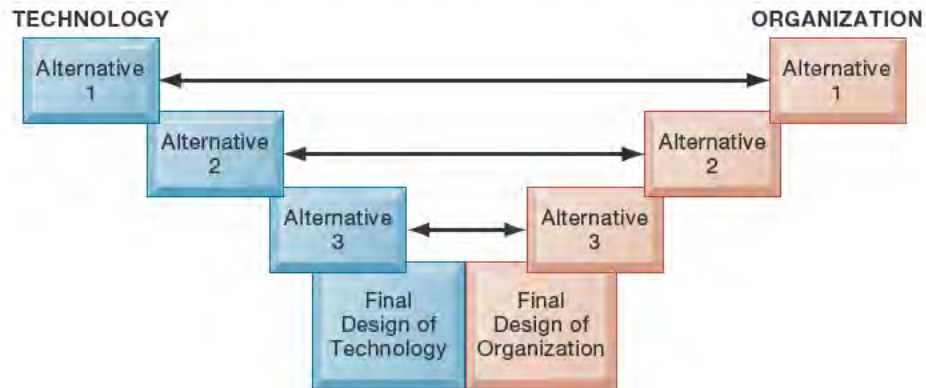
The view we adopt in this book is best characterized as the **sociotechnical view** of systems. In this view, optimal organizational performance is achieved by jointly optimizing both the social and technical systems used in production.

Adopting a sociotechnical systems perspective helps to avoid a purely technological approach to information systems. For instance, the fact that information technology is rapidly declining in cost and growing in power does not necessarily or easily translate into productivity enhancement or bottom-line profits. The fact that a firm has recently installed an enterprise-wide financial reporting system does not necessarily mean that it will be used, or used effectively. Likewise, the fact that a firm has recently introduced new business procedures and processes does not necessarily mean employees will be more productive in the absence of investments in new information systems to enable those processes.

In this book, we stress the need to optimize the firm's performance as a whole. Both the technical and behavioral components need attention. This means that technology must be changed and designed in such a way as to fit

FIGURE 1.10 A SOCIOTECHNICAL PERSPECTIVE ON INFORMATION SYSTEMS

In a sociotechnical perspective, the performance of a system is optimized when both the technology and the organization mutually adjust to one another until a satisfactory fit is obtained.



organizational and individual needs. Sometimes, the technology may have to be “de-optimized” to accomplish this fit. For instance, mobile phone users adapt this technology to their personal needs, and as a result manufacturers quickly seek to adjust the technology to conform to user expectations. Organizations and individuals must also be changed through training, learning, and planned organizational change to allow the technology to operate and prosper. Figure 1.10 illustrates this process of mutual adjustment in a sociotechnical system.



1-4 How will MIS help my career?

Here is how Chapter 1 and this text can help you find an entry-level job as a financial client support and sales assistant.

The Company

Power Financial Analytics Data Services, a data and software company serving the financial industry with offices in London, New York, Sydney, and Singapore, is looking to fill an entry-level position for a financial client support and sales assistant. The company has 1,600 employees, many of whom are consultants showing clients how to work with its powerful financial analytics software and data products.

Position Description

The financial client support and sales assistant will be part of a team in the company's consulting services. Consulting teams combine a thorough understanding of finance and technology with specific expertise in Power Financial Analytics Data Services software and assist clients in a variety of ways. The

company provides on-the-job training in its software and consulting methods.

Job responsibilities include:

- Supporting Financial Analytics Data Services applications.
- Helping the team create custom models and screens.
- Training clients in their offices and at seminars.
- Providing expert consultation to clients by telephone and on-site.

Job Requirements

- Recent college graduate or investment professional with one to two years of experience. Applicants with backgrounds in finance, MIS, economics, accounting, business administration, and mathematics are preferred
- Knowledge of or interest in learning about financial markets
- Sound working knowledge of spreadsheets
- Very strong communication and interpersonal skills
- Strong desire to learn in rapidly changing environment

Interview Questions

1. What is your background in finance? What courses did you take? Have you ever worked in the financial industry? What did you do there?
2. What is your proficiency level with spreadsheet software? What work have you done with Excel spreadsheets? Can you show examples of your work?
3. Are you able to discuss current trends in the financial industry and how they impact Power Financial's business model and client base?
4. Did you ever work with clients? Can you give examples of how you provided client service or support?
5. Can you give us an example of a finance-related problem or other business problem that you helped solve? Did you do any writing and analysis? Can you provide examples?

Author Tips

1. Use the web to learn about financial markets and the financial industry.
2. Use the web to research the company, its financial products, and the tools and services it offers customers. Learn what you can about its consulting services. Additionally, examine the company's social medial channels, such as LinkedIn and Facebook, for trends and themes.
3. Inquire exactly how you would be using spreadsheets for this job. Provide examples of how you used spreadsheets to solve problems in the classroom or for a job assignment. Show the spreadsheet work you did in finance.
4. Bring examples of your writing (including some from your Digital Portfolio described in MyLab MIS) demonstrating your analytical skills and project experience. Be prepared to discuss how you helped customers solve a business problem or the business problem solving you did for your courses.

REVIEW SUMMARY

1-1 How are information systems transforming business, and why are they so essential for running and managing a business today?

Organizations are trying to become more competitive and efficient by digitally enabling their core business processes and evolving into digital firms. The Internet has stimulated globalization by dramatically reducing the costs of producing, buying, and selling goods on a global scale. New information system trends include the emerging mobile digital platform, big data (including IoT), more remote management, democratization of decision making, machine learning systems, and the growing use of social media in business.

Information systems are a foundation for conducting business today. In many industries, survival and the ability to achieve strategic business goals are difficult without extensive use of information technology. Businesses today use information systems to achieve six major objectives: operational excellence; new products, services, and business models; customer/supplier intimacy; improved decision making; competitive advantage; and day-to-day survival.

1-2 What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?

From a technical perspective, an information system collects, stores, and disseminates information from an organization's environment and internal operations to support organizational functions and decision making, communication, coordination, control, analysis, and visualization. Information systems transform raw data into useful information through three basic activities: input, processing, and output.

From a business perspective, an information system provides a solution to a problem or challenge facing a firm and represents a combination of management, organization, and technology elements. The management dimension of information systems involves issues such as leadership, strategy, and management behavior. The technology dimension consists of computer hardware, software, data management technology, and networking/telecommunications technology (including the Internet). The organization dimension of information systems involves issues such as the organization's hierarchy, functional specialties, business processes, culture, and political interest groups.

In order to obtain meaningful value from information systems, organizations must support their technology investments with appropriate complementary investments in organizations and management. These complementary assets include new business models and business processes, supportive organizational culture and management behavior, and appropriate technology standards, regulations, and laws. New information technology investments are unlikely to produce high returns unless businesses make the appropriate managerial and organizational changes to support the technology.

1-3 What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

The study of information systems deals with issues and insights contributed from technical and behavioral disciplines. The disciplines that contribute to the technical approach focusing on formal models and capabilities of systems are computer science, management science, and operations research. The disciplines contributing to the behavioral approach focusing on the design, implementation, management, and business impact of systems are psychology, sociology, and economics. A sociotechnical view of systems considers both technical and social features of systems and solutions that represent the best fit between them.

Key Terms

- Business functions*, 49
Business model, 44
Business processes, 42
Complementary assets, 57
Computer hardware, 51
Computer literacy, 48
Computer software, 51
Culture, 50
Data, 46
Data management technology, 51
Data workers, 49
Digital firm, 42
Extranets, 52
Feedback, 47
Information, 46
Information system, 46
Information systems literacy, 48
Information technology (IT), 46
Information technology (IT) infrastructure, 52
Input, 47
Internet, 51
Intranets, 52
Knowledge workers, 49
Management information systems (MIS), 48
Middle management, 49
Network, 51
Networking and telecommunications technology, 51
Operational management, 49
Organizational and management capital, 57
Output, 47
Processing, 47
Production or service workers, 49
Senior management, 49
Sociotechnical view, 59
World Wide Web, 52

MyLab MIS

To complete the problems with MyLab MIS, go to the EOC Discussion Questions in MyLab MIS.

Review Questions

1-1 How are information systems transforming business, and why are they so essential for running and managing a business today?

- Describe how information systems have changed the way businesses operate and their products and services.
- Identify three major new information system trends.
- Describe the characteristics of a digital firm.
- Describe the challenges and opportunities of globalization in a “flattened” world.
- List and describe six reasons why information systems are so important for business today.

1-2 What is an information system? How does it work? What are its management, organization, and technology components? Why are complementary assets essential for ensuring that information systems provide genuine value for organizations?

- Explain the differences between information and data. Why is the former essential to the latter?
- Describe the three activities in an information system that produce the information that organizations need.

- Explain why organizations have a structure composed of different levels and specialties.
- Describe how the parts of an organization's culture can be found embedded in its information systems.
- Describe the features of an organization's information value chain.
- Based on what you have read, identify the key elements of organizational and management capital.

1-3 What academic disciplines are used to study information systems, and how does each contribute to an understanding of information systems?

- List and describe each discipline that contributes to a technical approach to information systems.
- List and describe each discipline that contributes to a behavioral approach to information systems.
- Describe the sociotechnical perspective on information systems.

Discussion Questions

1-4 MyLab MIS Information systems are too important to be left to computer specialists. Do you agree? Why or why not?

1-5 MyLab MIS If you were setting up the website for a professional football team, what management, organization, and technology issues might you encounter?

1-6 MyLab MIS What are some of the organizational, managerial, and social complementary assets that help make UPS's information systems so successful?

Hands-On MIS Projects

The projects in this section give you hands-on experience in analyzing financial reporting and inventory management problems, using data management software to improve management decision making about increasing sales, and using Internet software for researching job requirements. Visit MyLab MIS to access this chapter's Hands-On MIS Projects.

Management Decision Problems

- 1-7** Magical Toys is a South African toy store chain. The toys are sold in their five stores and from their own web shop. Each Tuesday, management needs sales reports from last week's activities. Work at the stores is hectic, so usually at least one store fails to deliver the report, making it impossible to sum up the previous week's sales. The reports, sent by email as spreadsheets, sometimes have inconclusive or erroneous data, and following up means that employees are unable to give their full focus to the customers. Some errors in the spreadsheets may also cause the purchasing manager to place unnecessary purchase orders. What are the main issues that Magical Toys faces in the absence of an information system? How could an information system be of help, and what issues can an information system not solve?
- 1-8** Diskonto operates deep-discount stores in Estonia offering housewares, cleaning supplies, clothing, health and beauty aids, and packaged food, with most items selling for a very low price. Its business model calls for keeping costs as low as possible. The company has no automated method for keeping track of inventory at each store. Managers know approximately how many cases of a particular product the store is supposed to receive when a delivery truck arrives, but the stores lack technology for scanning the cases or verifying the item count inside the cases. Merchandise losses from theft or other mishaps have been rising and now represent more than 3 percent of total sales. What decisions have to be made before investing in an information system solution?

Improving Decision Making: Using Databases to Analyze Sales Trends

Software skills: Database querying and reporting

Business skills: Sales trend analysis

- 1-9** In this project, you will start out with raw transactional sales data and use Microsoft Access database software to develop queries and reports that help managers make better decisions about product pricing, sales promotions, and inventory replenishment. In MyLab MIS, you can find a Store and Regional Sales Database developed in Microsoft Access. The database contains raw data on weekly store sales of computer equipment in various sales regions. The database includes fields for store identification number, sales region, item number, item description, unit price, units sold, and the weekly sales period when the sales were made. Use Access to develop some reports and queries to make this information more useful for running the business. Sales and production managers want answers to the following questions:
- Which products should be restocked?
 - Which stores and sales regions would benefit from a promotional campaign and additional marketing?
 - When (what time of year) should products be offered at full price, and when should discounts be used?
- You can easily modify the database table to find and report your answers. Print your reports and results of queries.

Improving Decision Making: Using the Internet to Locate Jobs Requiring Information Systems Knowledge

Software skills: Internet-based software

Business skills: Job searching

- 1-10** Visit a job-posting website such as EURES, the European Job Mobility Portal run by the European Commission. Spend some time at the site examining jobs for accounting, finance, sales, marketing, and human resources. Find two or three descriptions of jobs that require some information systems knowledge. What information systems knowledge do these jobs require? What do you need to do to prepare for these jobs? Write a one- to two-page report summarizing your findings.

Collaboration and Teamwork Project

Selecting Team Collaboration Tools

- 1-11** Form a team with three or four classmates and review the capabilities of Google Drive and Google Sites for your team collaboration work. Compare the capabilities of these two tools for storing team documents, project announcements, source materials, work assignments, illustrations, electronic presentations, and web pages of interest. Learn how each works with Google Docs. Explain why Google Drive or Google Sites is more appropriate for your team. If possible, use Google Docs to brainstorm and develop a presentation of your findings for the class. Organize and store your presentation using the Google tool you have selected.

Changes in the Financial Industry: Adyen and Fintech

CASE STUDY

Although 2020 was a very difficult year for many companies, it was a very successful one for Adyen, a Dutch company that mostly focuses on processing Internet payments. As the COVID-19 pandemic boosted the volume of online payments, business soared. Adyen announced in November 2020 that it was expanding in the Middle East, opening a new office in Dubai. Earlier that year, Internet retailer Zalando chose Adyen to process its credit card payments in all European markets. Raffles Hotel Singapore selected Adyen as its preferred payment provider to offer a seamless and more customer-centric digital payment experience across all of the hotel's touchpoints. The company also signed a new agreement with Microsoft, expanding and deepening their cooperation on payments processing. These are just the latest in a long series of successes for Adyen. In 2018, American Internet giant eBay announced that Adyen would become eBay's primary payments provider. This was significant for two reasons. First, eBay is, of course, a huge company: in 2019, the value of goods sold on the platform amounted to a staggering \$90.2 billion. For Adyen, this meant a huge increase of its business. Second, and perhaps more importantly, eBay gave preference to Adyen over PayPal, which is very often the first choice of online companies. For a startup like Adyen to beat a market leader like PayPal was remarkable.

As a result of all these successes, Adyen has been the star of the Amsterdam stock exchange for several years. In 2018, Adyen had a very successful initial public offering (IPO) and its shares, which were priced at €240, nearly doubled in price during the first trading day. Since then Adyen's share price has continued to increase, reaching over €740 by the end of 2019. Then the coronavirus crisis broke and more and more people started to shop online and make payments through Adyen. The company's share price more than doubled over the following months and by December 11, 2020, it had risen to €1834.50.

What explains Adyen's success? The answer to that rests primarily in the technology it uses. To put it simply, Adyen makes payment transactions easier. Making payments on the Internet is a much more complicated business than many people think it is,

and requires several companies to work together to ensure that the payment is successful. To start with, there is, of course, the merchant, but there are also companies providing a gateway for the payments, assessing the risk, and so on. Because of this crowded field of facilitators, many things can go wrong. For instance, software may be outdated, or programs used by one provider may be incompatible with those of another.

According to Forrester, a market research firm that focuses on information technology, around 10 percent of all payments fail for "technical" reasons. Adyen is reducing this performance gap by integrating several components of the payment chain (such as the gateway, risk assessment, and processing) into one platform. It has thus been successful in ensuring a higher success rate in its payments chain than other facilitators.

Adyen's second advantage is that its payments platform is connected to most payment systems being used in the world. Payment systems can differ greatly from country to country. In the Netherlands, for instance, most online payments are done through the iDEAL system, which links payments to checking accounts. In other countries, most payments are done through credit cards or debit cards like Visa or Mastercard. When it comes to processing payments, this variety of methods complicates things enormously, as the payment processors need to have several platforms. Adyen, however, has one integrated platform that can link with most payment methods being used all over the world. This is, of course, an attractive option for merchants or online companies that sell their goods worldwide. When eBay decided to work with Adyen, it gave two reasons for its decision: Adyen processes payments more cheaply and gives more control to merchants that use the platform.

Adyen is a good example of a company that works in the field of fintech (financial technology). Companies that use technology to facilitate financial transactions have been highly successful over the last decade. Founded in 2006, Adyen is relatively old for a fintech company, but it has grown enormously—in 2019 it had €240 billion in processed volume. Remarkable as Adyen's case is, it is definitely not unique; other firms have used technology to find a

new niche in the international payments universe. The company TransferWise (which was founded in Estonia but now mostly operates from the United Kingdom) is another example: it facilitates money transfers from Country A to Country B while avoiding currency conversion.

Initially, traditional banks found it hard to compete with companies like Adyen. It is in the DNA of fintech companies to develop and focus on state-of-the-art information systems. Traditional banks are very different; they tend to value direct, physical contact with the customer and often have branches in various countries that function more or less independently. Their IT staff often has a supporting role and not, as in the case of Adyen, a guiding one. Traditional banks use IT systems to facilitate business operations, not to compete with other financial providers. Their IT systems are usually at the national level and may differ from country to country. The heart of Adyen is its technical platform, and the IT staff works continuously to make it cheaper and more user-friendly. The data received on this platform are analyzed and used to upgrade and update it. The data that Adyen gets are global (the platform processes payments from all over the world) and highly diverse (from payment transactions on the Internet as well as transactions performed in physical shops). This wealth of data ensures that Adyen has better opportunities to analyze the market of payment transactions and its developments than traditional banks do.

Fintech is a good example of what economists call “creative destruction.” This term, originally coined by the Austrian economist Joseph Schumpeter, describes the influence of technology on the economy. According to his theory, new technologies inevitably result in winners and in losers. One famous example of a disruption is the Spinning Jenny, a device invented in Britain during the eighteenth century. The Spinning Jenny made the production of cloth more efficient, cheaper, and—critically—reduced the number of laborers needed in the textiles industry. Over the last 30 years, the Internet has had the same effect: some jobs have disappeared, and others have come into existence. A good example is travel agencies. Forty years ago, there was a travel agency on every high street, but today, the vast majority of travel is booked online. Likewise, traditional hotels now have to compete with sites like Airbnb.

Creative destruction comes in waves, and the banking industry is now in the frontline of change because of the emergence of fintech companies. Adyen has

incorporated many stages of the payment chain, and traditional companies that used to provide, for instance, risk assessment now have to either make their business model more efficient or face the very real possibility of going out of business. The German firm N26 is also a good example of how fintech is changing the banking industry. This company provides payment services just like a normal bank does, but it does not have physical locations where customer agents give advice to the customers; everything is online. In 2020, N26 had over 5 million customers in 25 markets. One of its founders, Maximilian Tayenthal, remarked that traditional banks had not taken N26 seriously when the company was launched. See the Chapter 3 opening case for more information on N26.

Fintech companies that make online banking much easier have already had a huge impact on the physical presence of banks in many countries. The number of bank branches in Europe has declined dramatically, from 240,000 a decade ago to around 165,000 in 2020. In the Netherlands, the digital banking experience progressed so rapidly that the Dutch government felt it necessary to make it clear that it would not allow the loss of cash payments from the Dutch High Streets as this would severely limit the participation in society of vulnerable groups such as the elderly.

According to experts, the next wave of technology will be even more disruptive. In 2018, Antony Jenkins, who ran Barclays from 2012 to 2015, told the BBC that in the coming years around 50 percent of all jobs in banking (including those of middle managers and customer agents) would be replaced by some kind of artificial intelligence. Andy Haldane, the chief economist of the Bank of England, warned in the same year of a fourth industrial revolution that would cause severe disruptions in the job market, and at the heart of it would be artificial intelligence—a modern counterpart to the Spinning Jenny.

This does not mean that success is automatically guaranteed for new companies like Adyen and TransferWise. Research shows that around 90 percent of new fintech companies go bankrupt without ever having been profitable. Many require large investments of capital before they can generate profits. Even when they generate profits, these are often, for a time at least, rather small. Fintech companies are now also being challenged by the major traditional banks, who are rapidly developing their own versions of online payment systems, from peer-to-peer payments for retail customers, to highly automated online payment and billing systems for global businesses.

As consumers and firms have moved towards mobile and online payments, many traditional banks have invested heavily in new technologies in part using the funds saved by closing down traditional physical bank branches. Where in-house innovation has not worked, the large banks have purchased fintech firms and folded them into their existing business and technology platforms. Traditional banks have a decided advantage over fintech startups: extraordinary cash flow and tens of millions of loyal customers and firms. In this scenario, fintech firms have been early to the party but typically do not survive to the end of the party, at least not as independent firms.

Adyen's founders were very experienced entrepreneurs when they started their company. They had already created another company, Bibit, which they sold to The Royal Bank of Scotland for €100 million. Indeed, the word adyen means "again" in a language from Surinam, meaning that Adyen is their second endeavor in the field of financial technology. Adyen appears likely to be one of the survivors. The volume of payments it processes has grown from €32 billion in 2015 to €240 billion in 2019. Adyen's revenues during this time period have also grown, from €98.5 billion in 2015 to almost €500 billion by 2019. Unlike many fintech company, Adyen has also been able to show a profit, with net income rising from €131 million in 2018 to €204 million in 2019, up 56 percent, and with

expectations that it will surpass all of those benchmarks for 2020. Adyen illustrates how fintech startups are changing the traditional bank and payment systems using new technologies and business models.

Sources: Bernard Vogelsang, "AEX Lager Het Weekend In Door Brexit-Zorgen," *Telegraaf*, December 11, 2020; "Adyen Breidt Samenwerking Met Microsoft Uit," *deaandeelhouder.nl*, December 10, 2020; Ayden, "Raffles Hotel Singapore Takes Its Legendary Guest Experience to the Next Level with Adyen," www.adyen.com, November 24, 2020; Robert Vink, "How Adyen Is Simultaneously Disrupting Banks and the Payments Industry," www.seekingalpha.com, November 18, 2020; Adyen, "Adyen to Expand in the Middle East, Opens Dubai Office," www.adyen.com, November 10, 2020; Ben Dummet, "A Little-Known Payments Provider Grows into One of Europe's Most Valuable Finance Firm," *Wall Street Journal*, November 16, 2020; "Quarter of Bank Branches Will Close in Coming Three Years," *Consultancy.uk*, September 14, 2020; Adyen, "Adyen to Process European Credit Card Payments for Zalando," www.adyen.com, August 13, 2020; Ryan Browne, "Fintech Firm Adyen Posts 34% Surge in First-Quarter Revenue as Coronavirus Bolsters Online Payments," www.cnbc.com, April 21, 2020; Adyen, "2019 Annual Report," www.adyen.com, March 24, 2020; eBay, "eBay Inc. Reports Fourth Quarter and Full Year 2019 Results," www.ebay.com, January 28, 2020; Tomasz Gryniewicz, "This Fintech Attracts 10,000 Customers a Day," www.netguru.com, December 4, 2019; Kamal Ahmed, "Bank of England Chief Economist Warns on AI Jobs Threat," *BBC News*, www.bbc.com, August 20, 2018; Jan Hammer, "Adyen: A Global Success Born in Amsterdam," *Index Ventures*, www.indexventures.com, June 13, 2018; "Bank Jobs Could Fall by 50% in 10 Years, Says Ex-Barclays Boss," *BBC News*, www.bbc.com, November 26, 2015.

CASE STUDY QUESTIONS

- 1-12** According to Adyen, the payments platform has two advantages compared to other payment platforms. Describe these advantages in your own words.
- 1-13** Many fintech companies are very valuable on the stock exchange even though they make little or no profit. Explain this situation. Why

do investors pay so much for shares of companies that hardly make any profits?

- 1-14** Explain the term "creative destruction." Is the German bank N26 destroying jobs or creating new jobs?
- 1-15** Explain why such a high percentage of fintech startups go bankrupt before making any profits at all.

Case contributed by Bernard Bouwman

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2

Global E-business and Collaboration

LEARNING OBJECTIVES

After reading this chapter, you will be able to answer the following questions:

- 2-1** What are business processes? How are they related to information systems?
- 2-2** How do systems serve the different management groups in a business, and how do systems that link the enterprise improve organizational performance?
- 2-3** Why are systems for collaboration and social business so important, and what technologies do they use?
- 2-4** What is the role of the information systems function in a business?
- 2-5** How will MIS help my career?

CHAPTER CASES

Enterprise Social Networking Transforms Sharp Corporation into a More Innovative Connected Organization
The City of Mississauga Goes Digital
Higher Education Redefined: Virtual Education at Ahlia University
Is Social Business Good Business?

VIDEO CASES

IS in Action: VisionX Lighting Grows with Business One
CEMEX: Becoming a Social Business

MyLab MIS

Discussion Questions: 2-5, 2-6, 2-7; Hands-On MIS Projects: 2-8, 2-9, 2-10, 2-11;
eText with Conceptual Animations

Enterprise Social Networking Transforms Sharp Corporation into a More Innovative Connected Organization

Sharp Corporation is a Japanese multinational corporation that manufactures and sells telecommunications equipment, electric and electronic equipment, and electronic components, including LCDs (liquid crystal displays), multifunction printing devices, calculators, radios, microwave ovens, and sensors. Sharp's business philosophy emphasizes that the company does not merely strive to increase business volume, but is also dedicated to using unique innovative technology to contribute to the culture and welfare of people throughout the world. Since 2016 Sharp has been a subsidiary of Taiwan-based Foxconn Group and employs more than 52,000 people worldwide. Collaboration and sharing information are essential for the company's ongoing innovation and business success.

Despite an impressive product lineup, Sharp ran into financial trouble as new Asian competitors challenged its consumer electronics business. The company's core electronic manufacturing service was based on razor-thin profit margins. Foxconn has been working with Sharp's management to diversify Sharp's business portfolio and reorganize the company to be more effective and profitable.

Sharp management believed that to strengthen its global business, the company needed to change the way it worked—both its business processes and its organizational culture. In the past employees had to follow decisions made from the top down. Directives flowed downward. Sharp needed to move to an organizational culture based on two-way dialogue where lower-level employees as well as upper management could play leadership roles. The company's younger employees were looking for systems where they could share ideas and opinions freely with each other.

Sharp implemented Microsoft Yammer for this purpose. Microsoft Yammer is an enterprise social networking platform for internal business uses, although it can also create external networks linking to suppliers, customers, and others outside the organization. Yammer enables employees to create groups to collaborate on projects and share and edit documents, and it includes a news feed to find out what's happening within the



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company. A People Directory provides a searchable database of contact information, skills, and expertise. Yammer can be accessed through the web, desktop, and mobile devices and can be integrated with other Microsoft tools such as SharePoint and Office 365 to make other applications more “social.” (SharePoint is Microsoft’s platform for collaboration, document sharing, and document management. Office 365 is Microsoft’s online service for its desktop productivity applications such as word processing, spreadsheet, electronic presentations, and data management.)

Sharp created a pilot Yammer program in February 2013, with employee adoption voluntary and employees allowed to write about hobbies and interests as well as work-related topics. Within months there were 6,000 adopters, and the number of Yammer users within the company has passed 10,000. Sharp wants to deploy Yammer to all domestic employees in Japan and eventually to overseas offices.

Yammer has improved the flow of information between senior management and rank-and-file employees. Communication among employees themselves has increased appreciably, with some conversations occurring among different departments and different regions. For example, smartphone developers are able to share opinions about functions and user-friendliness throughout the company and also learn about activities in other departments that they would not normally hear about by reading Yammer postings.

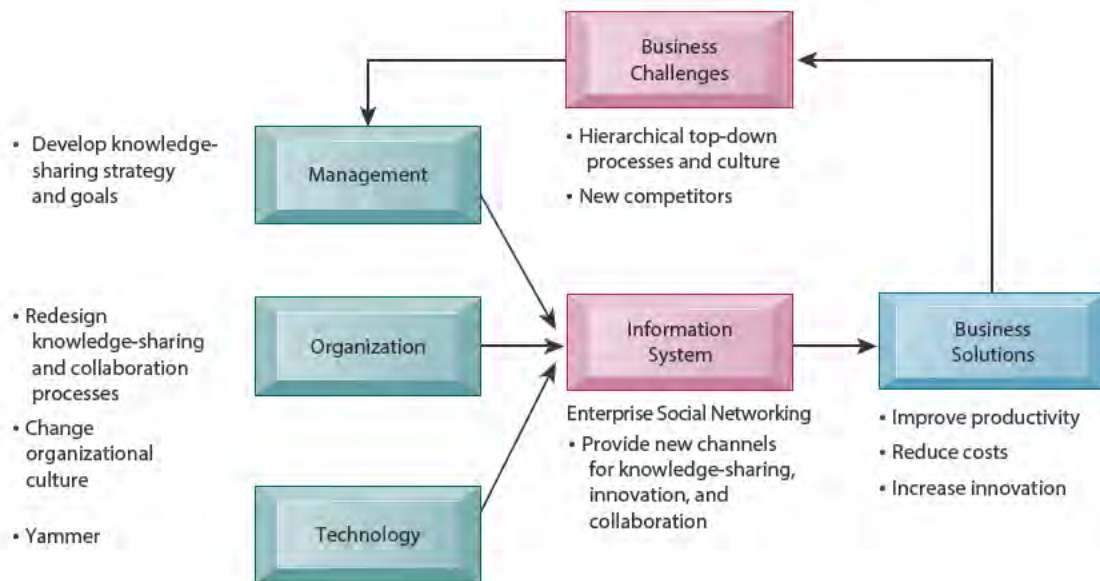
Some department leaders use Yammer to solicit ideas from their staff on how to use the latest technology in the business. Sharp is able to incorporate such feedback into product development and company policies. This ability to collect dispersed information from all over the company and have managers freely exchange ideas and learn from employees represents a major cultural change for the company.

Sources: “Sharp: A Culture Reborn Through Enterprise Social,” www.microsoft.com, accessed February 12, 2020; www.globalsharp.com, accessed January 21, 2020; and Sharp Corporation, “Sharp Is Back With Big Screens,” Hexus, February 5, 2019.

Sharp’s experience illustrates how much organizations today rely on information systems to improve their performance and remain competitive. It also shows how much systems supporting collaboration and teamwork make a difference in an organization’s ability to innovate, execute, grow profits, and remain competitive.

The chapter-opening diagram calls attention to important points raised by this case and this chapter. Sharp Corporation is a knowledge-intensive technology company that depends on innovation, but it was hampered by hierarchical top-down processes and a culture that prevented employees and managers from freely sharing information and innovating. This impacted the company’s ability to create and deliver new leading-edge products, maintain high profit margins, and fend off increasing competition from other companies.

Sharp management found that the best solution was to deploy new technology to help the company move from a hierarchical corporate knowledge and work environment to one that was more democratized. Enterprise social networking actively engaged employees and enabled them to obtain more knowledge from colleagues and managers. The company took advantage of Microsoft



Yammer's social tools to increase employee collaboration and engagement and to open up dialogues with management. There is now more effective sharing of employee knowledge, and the company has become more innovative and effective.

New technology alone would not have solved Sharp Corporation's problem. To make the solution effective, Sharp had to change its organizational culture and business processes for knowledge dissemination and collaborative work, and the new technology made these changes possible.

Here are some questions to think about: How are collaboration and employee engagement helping Sharp Corporation to be more competitive? How did using Yammer change corporate culture and the way work was performed at Sharp Corporation?

2-1 What are business processes? How are they related to information systems?

In order to operate, businesses must deal with many different pieces of information about suppliers, customers, employees, invoices, payments, and of course their products and services. They must organize work activities that use this information to operate efficiently and enhance the overall performance of the firm. Information systems make it possible for firms to manage all their information, make better decisions, and improve the execution of their business processes.

Business Processes

Business processes, which we introduced in Chapter 1, refer to the manner in which work is organized, coordinated, and focused to produce a valuable product or service. Business processes are the collection of activities required to produce a product or service. These activities are supported by flows of material, information, and knowledge among the participants in business processes. Business processes also refer to the unique ways in which organizations

coordinate work, information, and knowledge, and the ways in which management chooses to coordinate work.

To a large extent, the performance of a business firm depends on how well its business processes are designed and coordinated. A company's business processes can be a source of competitive strength if they enable the company to innovate or to execute better than its rivals. Business processes can also be liabilities if they are based on inefficient ways of working that impede organizational responsiveness and efficiency. The chapter-opening case describing Sharp Corporation's improvements in knowledge-sharing processes clearly illustrates these points, as do many of the other cases in this text.

Every business can be seen as a collection of business processes, some of which are part of larger encompassing processes. For instance, uses of mentoring, wikis, blogs, and videos are all part of the overall knowledge management process. Many business processes are tied to a specific functional area. For example, the sales and marketing function is responsible for identifying customers, and the human resources function is responsible for hiring employees. Table 2.1 describes some typical business processes for each of the functional areas of business.

Other business processes cross many different functional areas and require coordination across departments. For instance, consider the seemingly simple business process of fulfilling a customer order (see Figure 2.1). Initially, the sales department receives a sales order. The order passes first to accounting to ensure the customer can pay for the order either by a credit verification or request for immediate payment prior to shipping. Once the customer credit is established, the production department pulls the product from inventory or produces the product. Then the product is shipped (and this may require working with a logistics firm, such as UPS or FedEx). A bill or invoice is generated by the accounting department, and a notice is sent to the customer indicating that the product has shipped. The sales department is notified of the shipment and prepares to support the customer by answering calls or fulfilling warranty claims.

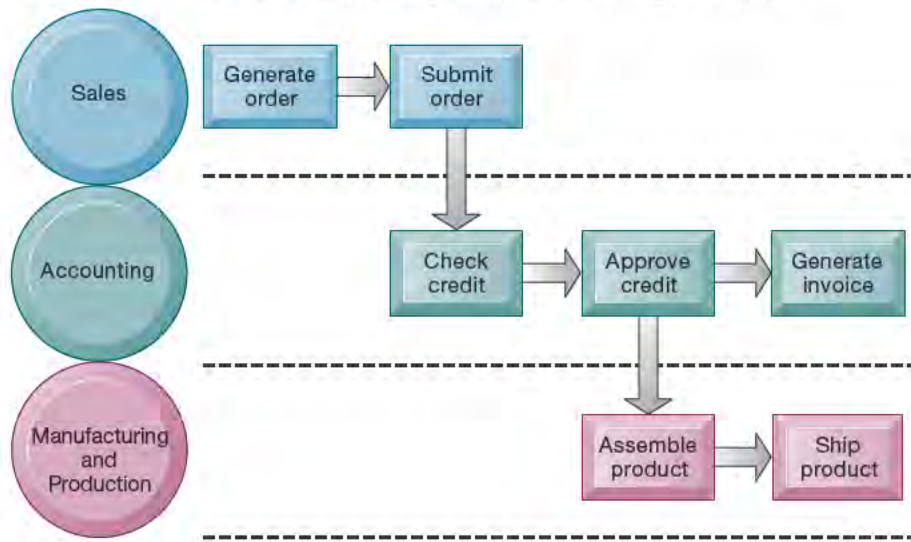
What at first appears to be a simple process, fulfilling an order, turns out to be a very complicated series of business processes that require the close

TABLE 2.1 **EXAMPLES OF FUNCTIONAL BUSINESS PROCESSES**

FUNCTIONAL AREA	BUSINESS PROCESS
Manufacturing and production	Assembling the product
	Checking for quality
	Producing bills of materials
Sales and marketing	Identifying customers
	Making customers aware of the product
	Selling the product
Finance and accounting	Paying creditors
	Creating financial statements
	Managing cash accounts
Human resources	Hiring employees
	Evaluating employees' job performance
	Enrolling employees in benefits plans

FIGURE 2.1 THE ORDER FULFILLMENT PROCESS

Fulfilling a customer order involves a complex set of steps that requires the close coordination of the sales, accounting, and manufacturing functions.



coordination of major functional groups in a firm. Moreover, to efficiently perform all these steps in the order fulfillment process requires a great deal of information. The required information must flow rapidly within the firm from one decision maker to another; with business partners, such as delivery firms; and with the customer. Computer-based information systems make this possible.

How Information Technology Improves Business Processes

Exactly how do information systems improve business processes? Information systems automate many steps in business processes that were formerly performed manually, such as checking a client's credit or generating an invoice and shipping order. But today, information technology can do much more. New technology can actually change the flow of information, making it possible for many more people to access and share information, replacing sequential steps with tasks that can be performed simultaneously, and eliminating delays in decision making. New information technology frequently changes the way a business works and supports entirely new business models. Downloading a Kindle e-book from Amazon, buying a computer online at Scan UK, and streaming a music track from Apple Music are entirely new business processes based on new business models that would be inconceivable without today's information technology.

That's why it's so important to pay close attention to business processes, both in your information systems course and in your future career. By analyzing business processes, you can achieve a very clear understanding of how a business actually works. Moreover, by conducting a business process analysis, you will also begin to understand how to change the business by improving its processes to make it more efficient or effective. Throughout this book, we examine business processes with a view to understanding how they might be improved by using information technology to achieve greater efficiency, innovation, and customer service.

2-2 How do systems serve the different management groups in a business, and how do systems that link the enterprise improve organizational performance?

Now that you understand business processes, it is time to look more closely at how information systems support the business processes of a firm. Because there are different interests, specialties, and levels in an organization, there are different kinds of systems. No single system can provide all the information an organization needs.

A typical business organization has systems supporting processes for each of the major business functions—sales and marketing, manufacturing and production, finance and accounting, and human resources. You can find examples of systems for each of these business functions in the Learning Tracks for this chapter. Functional systems that operate independently of each other are becoming a thing of the past because they cannot easily share information to support cross-functional business processes. Many have been replaced with large-scale cross-functional systems that integrate the activities of related business processes and organizational units. We describe these integrated cross-functional applications later in this section.

A typical firm also has different systems supporting the decision-making needs of each of the main management groups we described in Chapter 1. Operational management, middle management, and senior management each use systems to support the decisions they must make to run the company. Let's look at these systems and the types of decisions they support.

Systems for Different Management Groups

A business firm has systems to support different groups or levels of management. These systems include transaction processing systems and systems for business intelligence.

Transaction Processing Systems

Operational managers need systems that keep track of the elementary activities and transactions of the organization, such as sales, receipts, cash deposits, payroll, credit decisions, and the flow of materials in a factory. **Transaction processing systems (TPS)** provide this kind of information. A transaction processing system is a computerized system that performs and records the daily routine transactions necessary to conduct business, such as sales order entry, hotel reservations, payroll, employee record keeping, and shipping.

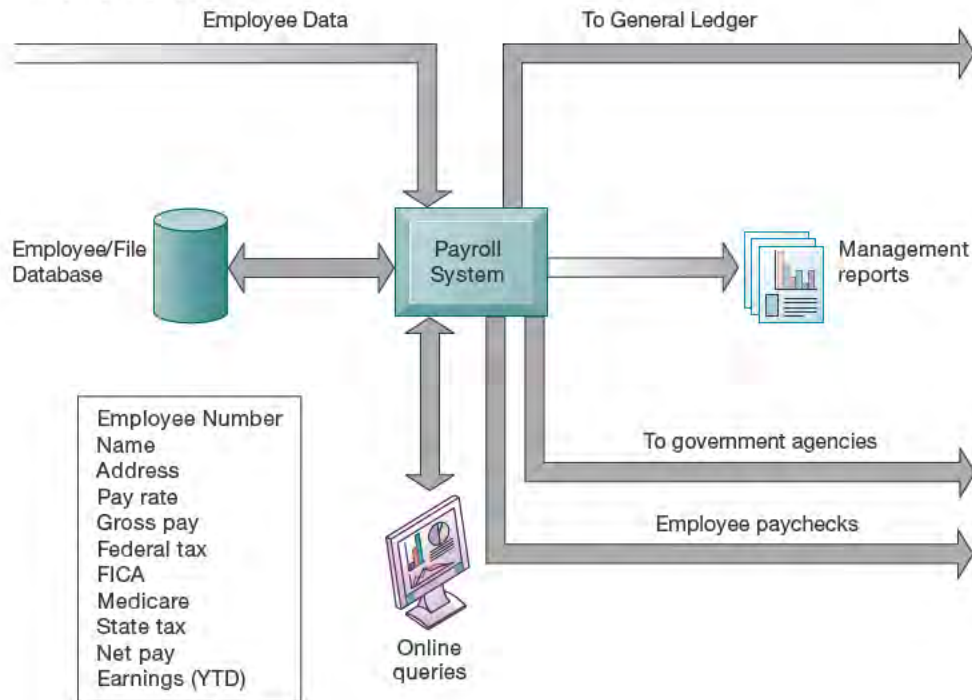
The principal purpose of systems at this level is to answer routine questions and to track the flow of transactions through the organization. How many parts are in inventory? What happened to Mr. Smith's payment? To answer these kinds of questions, information generally must be easily available, current, and accurate.

At the operational level, tasks, resources, and goals are predefined and highly structured. The decision to grant credit to a customer, for instance, is made by a lower-level supervisor according to predefined criteria. All that must be determined is whether the customer meets the criteria.

Figure 2.2 illustrates a TPS for payroll processing. A payroll system keeps track of money paid to employees. An employee time sheet with the employee's name, social security number, and number of hours worked per week

FIGURE 2.2 A PAYROLL TPS

A TPS for payroll processing captures employee payment transaction data (such as a time card). System outputs include online and hard-copy reports for management and employee paychecks.



represents a single transaction for this system. Once this transaction is input into the system, it updates the system's master file (or database—see Chapter 6) that permanently maintains employee information for the organization. The data in the system are combined in different ways to create reports of interest to management and government agencies and to send paychecks to employees.

Managers need TPS to monitor the status of internal operations and the firm's relations with the external environment. TPS are also major producers of information for the other systems and business functions. For example, the payroll system illustrated in Figure 2.2, along with other accounting TPS, supplies data to the company's general ledger system, which is responsible for maintaining records of the firm's income and expenses and for producing reports such as income statements and balance sheets. It also supplies employee payment history data for insurance, pension, and other benefits calculations to the firm's human resources function and employee payment data to government agencies.

Transaction processing systems are often so central to a business that TPS failure for a few hours can lead to a firm's demise and perhaps that of other firms linked to it. Imagine what would happen to UPS if its package tracking system was not working! What would the airlines do without their computerized reservation systems?

Systems for Business Intelligence

Firms also have business intelligence systems that focus on delivering information to support management decision making. **Business intelligence** is a contemporary term for data and software tools for organizing, analyzing, and

providing access to data to help managers and other enterprise users make more informed decisions. Business intelligence addresses the decision-making needs of all levels of management. This section provides a brief introduction to business intelligence. You'll learn more about this topic in Chapters 6 and 12.

Business intelligence systems for middle management help with monitoring, controlling, decision-making, and administrative activities. In Chapter 1, we defined management information systems as the study of information systems in business and management. The term **management information systems (MIS)** also designates a specific category of information systems serving middle management. MIS provide middle managers with reports on the organization's current performance. This information is used to monitor and control the business and predict future performance.

MIS summarize and report on the company's basic operations using data supplied by transaction processing systems. The basic transaction data from TPS are compressed and usually presented in reports that are produced on a regular schedule. Today, many of these reports are delivered online. Figure 2.3 shows how a typical MIS transforms transaction-level data from inventory, production, and accounting into MIS files that are used to provide managers with reports. Figure 2.4 shows a sample report from this system.

MIS typically provide answers to routine questions that have been specified in advance and have a predefined procedure for answering them. For instance, MIS reports might list the total pounds of lettuce used this quarter by a fast-food chain or, as illustrated in Figure 2.4, compare total annual sales figures for specific products to planned targets. These systems generally are not flexible and have little analytical capability. Most MIS use simple routines, such as summaries and comparisons, as opposed to sophisticated mathematical models or statistical techniques.

Other types of business intelligence systems support more nonroutine decision making. **Decision-support systems (DSS)** focus on problems that are unique and rapidly changing, for which the procedure for arriving at a solution may not be fully predefined in advance. They try to answer questions such as

FIGURE 2.3 HOW MANAGEMENT INFORMATION SYSTEMS OBTAIN THEIR DATA FROM THE ORGANIZATION'S TPS

In the system illustrated by this diagram, three TPS supply summarized transaction data to the MIS reporting system at the end of the time period. Managers gain access to the organizational data through the MIS, which provides them with the appropriate reports.

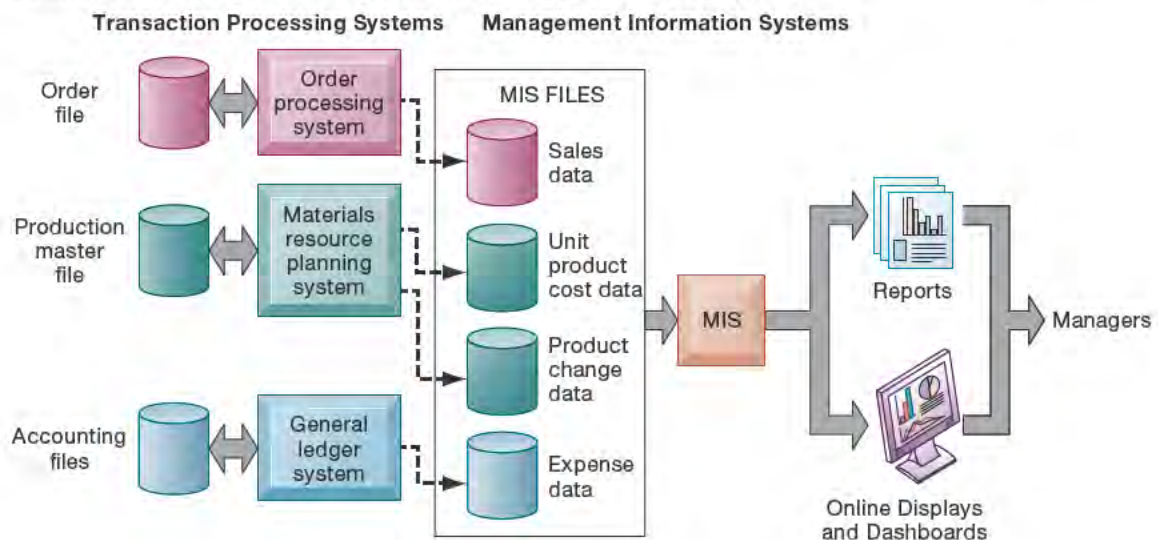


FIGURE 2.4 SAMPLE MIS REPORT

This report, showing summarized annual sales data, was produced by the MIS in Figure 2.3.

Consolidated Consumer Products Corporation Sales by Product and Sales Region: 2020

PRODUCT CODE	PRODUCT DESCRIPTION	SALES REGION	ACTUAL SALES	PLANNED	ACTUAL versus PLANNED
4469	Carpet Cleaner	Northeast	4,066,700	4,800,000	0.85
		South	3,778,112	3,750,000	1.01
		Midwest	4,867,001	4,600,000	1.06
		West	4,003,440	4,400,000	0.91
		TOTAL	16,715,253	17,550,000	0.95
5674	Room Freshener	Northeast	3,676,700	3,900,000	0.94
		South	5,608,112	4,700,000	1.19
		Midwest	4,711,001	4,200,000	1.12
		West	4,563,440	4,900,000	0.93
		TOTAL	18,559,253	17,700,000	1.05

these: What would be the impact on production schedules if we were to double sales in the month of December? What would happen to our return on investment if a factory schedule were delayed for six months?

Although DSS use internal information from TPS and MIS, they often bring in information from external sources, such as current stock prices or product prices of competitors. These systems are employed by “super-user” managers and business analysts who want to use sophisticated analytics and models to analyze data.

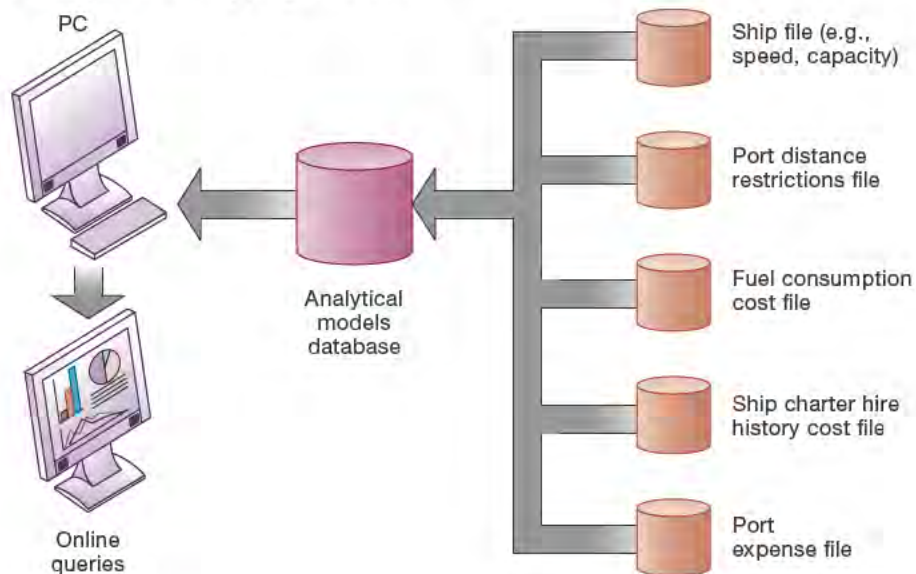
An interesting, small, but powerful DSS is the voyage-estimating system of a large global shipping company that transports bulk cargoes of coal, oil, ores, and finished products. The firm owns some vessels, charters others, and bids for shipping contracts in the open market to carry general cargo. A voyage-estimating system calculates financial and technical voyage details. Financial calculations include ship/time costs (fuel, labor, capital), freight rates for various types of cargo, and port expenses. Technical details include myriad factors, such as ship cargo capacity, speed, port distances, fuel and water consumption, and loading patterns (location of cargo for different ports).

The system can answer questions such as the following: Given a customer delivery schedule and an offered freight rate, which vessel should be assigned at what rate to maximize profits? What is the optimal speed at which a particular vessel can optimize its profit and still meet its delivery schedule? What is the optimal loading pattern for a ship bound for the U.S. West Coast from Malaysia? Figure 2.5 illustrates the DSS built for this company. The system operates on a powerful desktop personal computer, providing a system of menus that makes it easy for users to enter data or obtain information.

The voyage-estimating DSS we have just described draws heavily on models. Other business intelligence systems are more data-driven, focusing instead on extracting useful information from very large quantities of data. For example, large ski resort companies such as Intrawest and Vail Resorts collect and store large amounts of customer data from call centers, lodging and dining reservations, ski schools, and ski equipment rental stores. They use special software to analyze these data to determine the value, revenue potential, and loyalty of each customer to help managers make better decisions about how to target their marketing programs.

FIGURE 2.5 VOYAGE-ESTIMATING DECISION-SUPPORT SYSTEM

This DSS operates on a powerful PC. It is used daily by managers who must develop bids on shipping contracts.

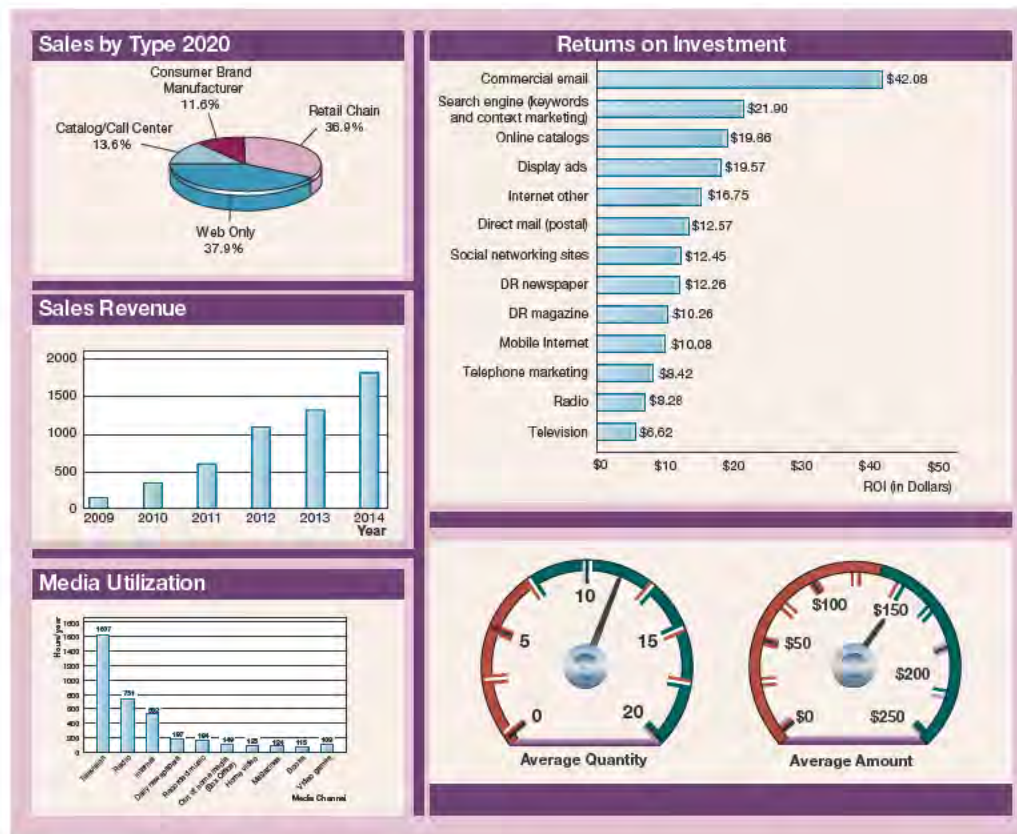


Business intelligence systems also address the decision-making needs of senior management. Senior managers need systems that focus on strategic issues and long-term trends, both in the firm and in the external environment. They are concerned with questions such as: What will employment levels be in five years? What are the long-term industry cost trends? What products should we be making in five years?

Executive support systems (ESS) help senior management make these decisions. They address nonroutine decisions requiring judgment, evaluation, and insight because there is no agreed-on procedure for arriving at a solution. ESS present graphs and data from many sources through an interface that is easy for senior managers to use. Often the information is delivered to senior executives through a **portal**, which uses a web interface to present integrated personalized business content.

ESS are designed to incorporate data about external events, such as new tax laws or competitors, but they also draw summarized information from internal MIS and DSS. They filter, compress, and track critical data, displaying the data of greatest importance to senior managers. Increasingly, such systems include business intelligence analytics for analyzing trends, forecasting, and "drilling down" to data at greater levels of detail.

For example, the chief operating officer (COO) and plant managers at Valero, the world's largest independent petroleum refiner, use a Refining Dashboard to display real-time data related to plant and equipment reliability, inventory management, safety, and energy consumption. With the displayed information, the COO and his team can review the performance of each Valero refinery in the United States and Canada in terms of how each plant is performing compared to the production plan of the firm. The headquarters group can drill down from executive level to refinery level and individual system-operator level displays of performance. Valero's Refining Dashboard is an example of a **digital dashboard**, which displays on a single screen graphs and charts of key performance indicators for managing a company. Digital dashboards are becoming an increasingly popular tool for management decision makers.



A digital dashboard delivers comprehensive and accurate information for decision making, often using a single screen. The graphical overview of key performance indicators helps managers quickly spot areas that need attention.

Systems for Linking the Enterprise

Reviewing all the different types of systems we have just described, you might wonder how a business can manage all the information in these different systems. You might also wonder how costly it is to maintain so many different systems. And you might wonder how all these different systems can share information and how managers and employees are able to coordinate their work. In fact, these are all important questions for businesses today.

Enterprise Applications

Getting all the different kinds of systems in a company to work together has proven a major challenge. Typically, corporations are put together both through normal “organic” growth and through acquisition of smaller firms. Over a period of time, corporations end up with a collection of systems, most of them older, and face the challenge of getting them all to “talk” with one another and work together as one corporate system. There are several solutions to this problem.

One solution is to implement **enterprise applications**, which are systems that span functional areas, focus on executing business processes across the firm, and include all levels of management. Enterprise applications help businesses become more flexible and productive by coordinating their business processes more closely and integrating groups of processes so they focus on efficient management of resources and customer service.

There are four major enterprise applications: enterprise systems, supply chain management systems, customer relationship management systems, and knowledge management systems. Each of these enterprise applications

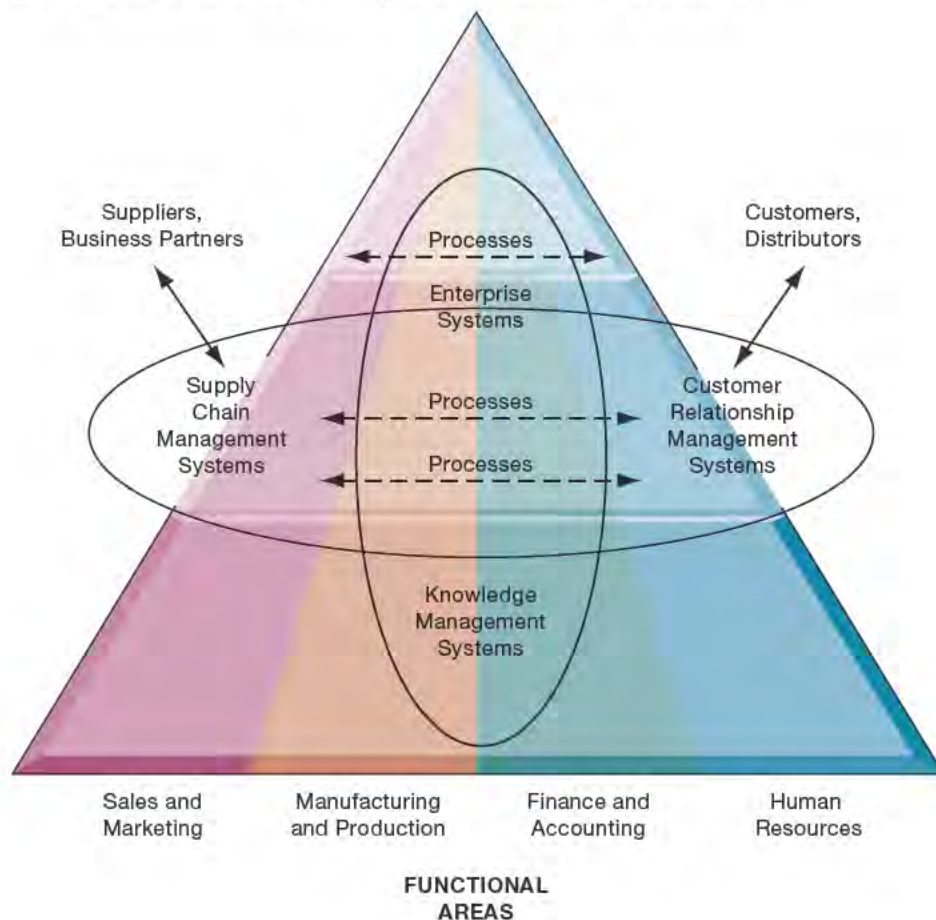
integrates a related set of functions and business processes to enhance the performance of the organization as a whole. Figure 2.6 shows that the architecture for these enterprise applications encompasses processes spanning the entire organization and, in some cases, extending beyond the organization to customers, suppliers, and other key business partners.

Enterprise Systems Firms use **enterprise systems**, also known as enterprise resource planning (ERP) systems, to integrate business processes in manufacturing and production, finance and accounting, sales and marketing, and human resources into a single software system. Information that was previously fragmented in many different systems is stored in a single comprehensive data repository where it can be used by many different parts of the business.

For example, when a customer places an order, the order data flow automatically to other parts of the company that are affected by them. The order transaction triggers the warehouse to pick the ordered products and schedule shipment. The warehouse informs the factory to replenish whatever has been depleted. The accounting department is notified to send the customer an invoice. Customer service representatives track the progress of the order through every step to inform customers about the status of their orders. Managers are able to use firmwide information to make more-precise and timely decisions about daily operations and longer-term planning.

FIGURE 2.6 ENTERPRISE APPLICATION ARCHITECTURE

Enterprise applications automate processes that span multiple business functions and organizational levels and may extend outside the organization.



Supply Chain Management Systems Firms use **supply chain management (SCM) systems** to help manage relationships with their suppliers. These systems help suppliers, purchasing firms, distributors, and logistics companies share information about orders, production, inventory levels, and delivery of products and services so they can source, produce, and deliver goods and services efficiently. The ultimate objective is to get the right amount of their products from their source to their point of consumption in the least amount of time and at the lowest cost. These systems increase firm profitability by lowering the costs of moving and making products and by enabling managers to make better decisions about how to organize and schedule sourcing, production, and distribution.

Supply chain management systems are one type of **interorganizational system** because they automate the flow of information across organizational boundaries. You will find examples of other types of interorganizational information systems throughout this text because such systems make it possible for firms to link digitally to customers and to outsource their work to other companies.

Customer Relationship Management Systems Firms use **customer relationship management (CRM) systems** to help manage their relationships with their customers. CRM systems provide information to coordinate all of the business processes that deal with customers in sales, marketing, and service to optimize revenue, customer satisfaction, and customer retention. This information helps firms identify, attract, and retain the most profitable customers; provide better service to existing customers; and increase sales.

Knowledge Management Systems Some firms perform better than others because they have better knowledge about how to create, produce, and deliver products and services. This firm knowledge is unique, is difficult to imitate, and can be leveraged into long-term strategic benefits. **Knowledge management systems (KMS)** enable organizations to better manage processes for capturing and applying knowledge and expertise. These systems collect all relevant knowledge and experience in the firm and make it available wherever and whenever it is needed to improve business processes and management decisions. They also link the firm to external sources of knowledge. We examine enterprise systems and systems for supply chain management and customer relationship management in greater detail in Chapter 9. We discuss collaboration systems that support knowledge management in this chapter and cover other types of knowledge management applications in Chapter 11.

Intranets and Extranets

Enterprise applications create deep-seated changes in the way the firm conducts its business, offering many opportunities to integrate important business data into a single system. They are often costly and difficult to implement. Intranets and extranets deserve mention here as alternative tools for increasing integration and expediting the flow of information within the firm and with customers and suppliers.

Intranets are simply internal company websites that are accessible only by employees. The term *intranet* refers to an internal network, in contrast to the Internet, which is a public network linking organizations and other external networks. Intranets use the same technologies and techniques as the larger Internet, and they often are simply a private access area in a larger company website. Likewise with extranets, which are company websites that are accessible to authorized vendors and suppliers and are often used to coordinate the movement of supplies to the firm's production apparatus.

For example, Six Flags, which operates 18 theme parks throughout North America, maintains an intranet for its 1,900 full-time employees that provides company-related news and information on each park's day-to-day operations,

including weather forecasts, performance schedules, and details about groups and celebrities visiting the parks. The company also uses an extranet to broadcast information about schedule changes and park events to its 30,000 seasonal employees. We describe the technology for intranets and extranets in more detail in Chapter 7.

E-business, E-commerce, and E-government

The systems and technologies we have just described are transforming firms' relationships with customers, employees, suppliers, and logistic partners into digital relationships using networks and the Internet. So much business is now enabled by or based upon digital networks that we use the terms *electronic business* and *electronic commerce* frequently throughout this text.

Electronic business, or e-business, refers to the use of digital technology and the Internet to execute the major business processes in the enterprise. E-business includes activities for the internal management of the firm and for coordination with suppliers and other business partners. It also includes **electronic commerce, or e-commerce**.

E-commerce is the part of e-business that deals with the buying and selling of goods and services over the Internet. It also encompasses activities supporting those market transactions, such as advertising, marketing, customer support, security, delivery, and payment.

The technologies associated with e-business have also brought about similar changes in the public sector. Governments on all levels are using Internet technology to deliver information and services to citizens, employees, and businesses with which they work. **E-government** refers to the application of the Internet and networking technologies to digitally enable government and public sector agencies' relationships with citizens, businesses, and other arms of government.

In addition to improving delivery of government services, e-government makes government operations more efficient and also empowers citizens by giving them easier access to information and the ability to network electronically with other citizens. For example, citizens in some states can renew their driver's licenses or apply for unemployment benefits online, and the Internet has become a powerful tool for instantly mobilizing interest groups for political action and fundraising. The Interactive Session on Organizations describes the Canadian City of Mississauga's e-government program.

INTERACTIVE SESSION ORGANIZATIONS

The City of Mississauga Goes Digital

The City of Mississauga, a Toronto suburb, is Canada's sixth-largest city and a leading user of digital technology to improve its operations and services. It tries to integrate technology into its operations and strategic and business planning, with technology roadmaps for each municipal service defined in business plans and budgets.

Mississauga has a vibrant multicultural population and a thriving central business district, with many Canadian and multinational corporations headquartered there. Since 1970, however, the Greater Toronto Area, including Mississauga, has experienced a noticeable increase in low-income families and a similar decrease in middle-income families. Mississauga developed a Smart City Master Plan to provide a vision and framework to guide the city's adoption of digital technology. The city's leaders believe that digital technology should be available for everyone and provide opportunities to startups, schools, and households at risk.

Mississauga has been a leader in technology initiatives such as social media and "bring your own device" (BYOD), which allows employees to use their own mobile devices for their jobs. Mississauga's website and online services are hosted in remote cloud computing centers accessible via the Internet. (Chapter 5 of this text provides a detailed discussion of these technologies.) In deploying these technologies, the city tries to focus on usability, a high-quality customer experience, and making information technology and technology services available to residents of all income and educational levels.

Mississauga is trying as much as possible to go paperless, with meetings and collaboration via videoconferencing where participants can "attend" meetings and share documents remotely. These efforts have significantly reduced paper use and the need to travel via car or airplane to meetings.

Mobile tools have made it possible for city staff, including transit operators and work operations staff in the field, who previously lacked computers, to access employee information and operational data to support real-time operations and decision making. Working with cellular provider Rogers Wireless, Mississauga connected over 600 buses that are collecting information about bus operations and routes,

so that the public has real-time information about bus locations. The bus data collected also are used for timing maintenance, warranty, and mileage routines so that buses can be removed from service at optimal times to minimize service interruption.

Mississauga has additionally connected 700 city vehicles such as fire trucks and vehicles for snow, public works, and parks operations, and facility maintenance to provide real-time location-based information. For example, connected snow vehicles provide real-time snow plow information to the public as well as the expected level of service for snow removal. Onboard sensors track when snow blades are active, where and when salt or sand is applied, and the rate at which these materials are applied. Mississauga recently implemented an Advanced Traffic Management System (ATMS), which connected over 700 traffic intersections using its own high-capacity fiber optic and wireless Wi-Fi networks (see Chapter 7) along with the Rogers cellular network.

A City Hall pilot project created individual workspaces and collaboration units on the fifth floor so that staff can choose where they want to sit and work. Over 90 percent of the staff on that floor have no defined desk or desk phone, but they do have mobile technology to connect anywhere. A mobile working environment has helped the city attract younger employees, and it has transformed the way top management works as well. The city manager, commissioners, and directors are trading desktops for mobile devices.

Partnering with three other municipalities, Mississauga built its own high-speed fiber optic network known as the Public Sector Network (PSN). It is the largest municipally owned fiber optic network in Canada. This supports a citywide high-speed fiber network for transmitting large quantities of data and a wireless Wi-Fi network that supplies wireless connectivity to the public for many city services. Enterprise networking giant Cisco Canada helped the city build an extensive Wi-Fi network for all its community centers, libraries, arenas, and many outdoor locations such as parks and small business areas. This free Wi-Fi network is available as a "virtual campus" to college and university students around

the world. In 2018 over 8 million hours of free public Wi-Fi were used across the City. Providing public Wi-Fi access in so many locations across the city is one way for Mississauga to level the “digital divide” between residents who are technology “haves” and those who are technology “have-nots.”

Mississauga is working with the United Way, Region of Peel, University of Toronto at Mississauga, Sheridan College, and its Business Improvement Areas (BOAs) to build a mobile-friendly ecosystem across the city that can deliver services and digital technology to the entire community. The plan divides Mississauga into 23 defined communities, with one Hub center and 500 mobility kits to residents enrolled in social support programs per community. Each mobility kit consists of a connected laptop. Hubs will be developed jointly with several of the

large technology firms with Canadian headquarters in Mississauga and they will provide coworking spaces where their employees can do their work. Eventually the city will have 100 Hubs. The city is also planning to build 500 “Connects” across its 23 communities that will provide indoor and outdoor spaces with voice-supported digital screens and free Wi-Fi access. A “Connect” could be in a park, beside a bus stop, or inside a mall, and there citizens will find free Wi-Fi, a place to sit, and access to services and programs.

Sources: “SMRTCTY Master Plan,” smartcity.mississauga.ca, accessed February 9, 2020; Sophie Chapman, “Inside the City of Mississauga’s Technology Transformation Journey,” Gigabit, February 18, 2019; and Eric Emin Wood, “How the City of Mississauga Uses Mobile Technology to Engage Workers and Citizens Alike,” IT World Canada, May 7, 2018.

CASE STUDY QUESTIONS

1. Describe the problems the City of Mississauga hoped to address using digital technology.
2. What technologies did Mississauga employ for a solution? Describe each of these technologies and the role each played in a solution.
3. What management, organization, and technology issues did the City of Mississauga have to address in developing a solution?
4. How did the technologies in this case improve operations and decision making at the City of Mississauga?

2-3 Why are systems for collaboration and social business so important, and what technologies do they use?

With all these systems and information, you might wonder how it is possible to make sense of them. How do people working in firms pull it all together, work toward common goals, and coordinate plans and actions? In addition to the types of systems we have just described, businesses need special systems to support collaboration and teamwork.

What Is Collaboration?

Collaboration is working with others to achieve shared and explicit goals. Collaboration focuses on task or mission accomplishment and usually takes place in a business or other organization and between businesses. You collaborate with a colleague in Tokyo who has expertise on a topic about which you know nothing. You collaborate with many colleagues in publishing a company blog. If you’re in a law firm, you collaborate with accountants in an accounting firm in servicing the needs of a client with tax problems.

Collaboration can be short-lived, lasting a few minutes, or longer term, depending on the nature of the task and the relationship among participants. Collaboration can be one-to-one or many-to-many.

Employees may collaborate in informal groups that are not a formal part of the business firm's organizational structure, or they may be organized into formal teams. **Teams** have a specific mission that someone in the business assigned to them. Team members need to collaborate on the accomplishment of specific tasks and collectively achieve the team mission. The team mission might be to "win the game" or "increase online sales by 10 percent." Teams are often short-lived, depending on the problems they tackle and the length of time needed to find a solution and accomplish the mission.

Collaboration and teamwork are more important today than ever for a variety of reasons.

- *Changing nature of work.* The nature of work has changed from factory manufacturing and precomputer office work where each stage in the production process occurred independently of one another and was coordinated by supervisors. Work was organized into silos. Within a silo, work passed from one machine tool station to another, from one desktop to another, until the finished product was completed. Today, jobs require much closer coordination and interaction among the parties involved in producing the service or product. For instance, a report from the consulting firm McKinsey & Company estimated that 41 percent of the U.S. labor force is now composed of jobs where interaction (talking, emailing, presenting, and persuading) is the primary value-adding activity (McKinsey Global Institute, 2012). Even in factories, workers today often work in production groups, or pods.
- *Growth of professional work.* "Interaction" jobs tend to be professional jobs in the service sector that require close coordination and collaboration. Professional jobs require substantial education and the sharing of information and opinions to get work done. Each actor on the job brings specialized expertise to the problem, and all the actors need to take one another into account in order to accomplish the job.
- *Changing organization of the firm.* For most of the industrial age, managers organized work in a hierarchical fashion. Orders came down the hierarchy, and responses moved back up the hierarchy. Today, work is organized into groups and teams, and the members are expected to develop their own methods for accomplishing the task. Senior managers observe and measure results but are much less likely to issue detailed orders or operating procedures. In part, this is because expertise and decision-making power have been pushed down in organizations.
- *Changing scope of the firm.* The work of the firm has changed from a single location to multiple locations—offices or factories throughout a region, a nation, or even around the globe. For instance, Henry Ford developed the first mass-production automobile plant at a single Dearborn, Michigan factory. In 2020, Ford employed 199,000 people at 61 plants worldwide. With this kind of global presence, the need for close coordination of design, production, marketing, distribution, and service obviously takes on new importance and scale. Large global companies need to have teams working on a global basis.
- *Emphasis on innovation.* Although we tend to attribute innovations in business and science to great individuals, these great individuals are most likely working with a team of brilliant colleagues. Think of Bill Gates and Steve Jobs (founders of Microsoft and Apple), both of whom are highly regarded innovators and both of whom built strong collaborative teams to nurture and support innovation in their firms. Their initial innovations derived from close collaboration with colleagues and partners. Innovation, in other words, is a group and social process, and most innovations derive from collaboration among individuals in a lab, a business, or government agencies. Strong collaborative practices and technologies are believed to increase the rate and quality of innovation.

- *Changing culture of work and business.* Most research on collaboration supports the notion that diverse teams produce better outputs faster than individuals working on their own. Popular notions of the crowd (“crowdsourcing” and the “wisdom of crowds”) also provide cultural support for collaboration and teamwork.

What Is Social Business?

Many firms today enhance collaboration by embracing **social business**—the use of social networking platforms, including Facebook, Twitter, and internal corporate social tools—to engage their employees, customers, and suppliers. These tools enable workers to set up profiles, form groups, and “follow” each other’s status updates. The goal of social business is to deepen interactions with groups inside and outside the firm to expedite and enhance information sharing, innovation, and decision making.

A key word in social business is *conversations*. Customers, suppliers, employees, managers, and even oversight agencies continually have conversations about firms, often without the knowledge of the firm or its key actors (employees and managers).

Supporters of social business argue that if firms could tune in to these conversations, they would strengthen their bonds with consumers, suppliers, and employees, increasing their emotional involvement in the firm.

All of this requires a great deal of information transparency. People need to share opinions and facts with others quite directly, without intervention from executives or others. Employees get to know directly what customers and other employees think, suppliers will learn very directly the opinions of supply chain partners, and even managers presumably will learn more directly from their employees how well they are doing. Nearly everyone involved in the creation of value will know much more about everyone else.

If such an environment could be created, it is likely to drive operational efficiencies, spur innovation, and accelerate decision making. If product designers can learn directly about how their products are doing in the market in real time, based on consumer feedback, they can speed up the redesign process. If employees can use social connections inside and outside the company to capture new knowledge and insights, they will be able to work more efficiently and solve more business problems.

Table 2.2 describes important applications of social business inside and outside the firm. This chapter focuses on enterprise social business—its internal corporate uses. Chapters 7 and 10 describe social business applications relating to customers and suppliers outside the company.

TABLE 2.2 APPLICATIONS OF SOCIAL BUSINESS

SOCIAL BUSINESS APPLICATION	DESCRIPTION
Social networks	Connect through personal and business profiles
Crowdsourcing	Harness collective knowledge to generate new ideas and solutions
Shared workspaces	Coordinate projects and tasks; cocreate content
Blogs and wikis	Publish and rapidly access knowledge; discuss opinions and experiences
Social commerce	Share opinions about purchasing on social platforms
File sharing	Upload, share, and comment on photos, videos, audio, text documents
Social marketing	Use social media to interact with customers; derive customer insights
Communities	Discuss topics in open forums; share expertise

Business Benefits of Collaboration and Social Business

Much of the research on collaboration has been anecdotal, but there is a general belief among both business and academic communities that the more a business firm is “collaborative,” the more successful it will be, and that collaboration within and among firms is more essential than in the past. *MIT Sloan Management Review*’s research found that a focus on collaboration is central to how digitally advanced companies create business value and establish competitive advantage (Kiron, 2017). A global survey of business and information systems managers found that investments in collaboration technology produced organizational improvements that returned more than four times the amount of the investment, with the greatest benefits for sales, marketing, and research and development functions (Frost & Sullivan, 2009). McKinsey & Company consultants predict that social technologies used within and across enterprises could potentially raise the productivity of interaction workers by 20 to 25 percent (McKinsey Global Institute, 2012).

Table 2.3 summarizes some of the benefits of collaboration and social business that have been identified. Figure 2.7 graphically illustrates how collaboration is believed to affect business performance.

Building a Collaborative Culture and Business Processes

Collaboration won’t take place spontaneously in a business firm, especially in the absence of supportive culture or business processes. Business firms, especially large firms, had a reputation in the past for being “command and control” organizations where the top leaders thought up all the really important matters and then ordered lower-level employees to execute senior management plans. The job of middle management supposedly was to pass messages back and forth, up and down the hierarchy.

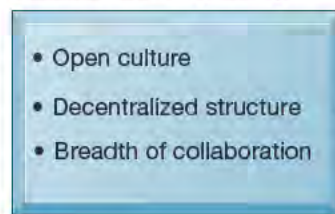
Command and control firms required lower-level employees to carry out orders without asking too many questions, with no responsibility to improve processes, and with no rewards for teamwork or team performance. If your work group needed help from another work group, that was something for the bosses to figure out. You never communicated horizontally, always vertically, so management could control the process. Together, the expectations of management

TABLE 2.3 BUSINESS BENEFITS OF COLLABORATION AND SOCIAL BUSINESS

BENEFIT	RATIONALE
Productivity	People interacting and working together can capture expert knowledge and solve problems more rapidly than the same number of people working in isolation from one another. There will be fewer errors.
Quality	People working collaboratively can communicate errors and corrective actions faster than if they work in isolation. Collaborative and social technologies help reduce time delays in design and production.
Innovation	People working collaboratively can come up with more innovative ideas for products, services, and administration than the same number working in isolation from one another. There are advantages to diversity and the “wisdom of crowds.”
Customer service	People working together using collaboration and social tools can solve customer complaints and issues faster and more effectively than if they were working in isolation from one another.
Financial performance (profitability, sales, and sales growth)	As a result of all of the above, collaborative firms have superior sales, sales growth, and financial performance.

FIGURE 2.7 REQUIREMENTS FOR COLLABORATION

Successful collaboration requires an appropriate organizational structure and culture along with appropriate collaboration technology.

Collaboration Capability**Collaboration Technology**

Collaboration
Quality

Firm Performance

and employees formed a culture, a set of assumptions about common goals and how people should behave. Many business firms still operate this way.

A collaborative business culture and business processes are very different. Senior managers are responsible for achieving results but rely on teams of employees to achieve and implement the results. Policies, products, designs, processes, and systems are much more dependent on teams at all levels of the organization to devise, to create, and to build. Teams are rewarded for their performance, and individuals are rewarded for their performance in a team. The function of middle managers is to build the teams, coordinate their work, and monitor their performance. The business culture and business processes are more “social.” In a collaborative culture, senior management establishes collaboration and teamwork as vital to the organization, and it actually implements collaboration for the senior ranks of the business as well.

Tools and Technologies for Collaboration and Social Business

A collaborative, team-oriented culture won't produce benefits without information systems in place to enable collaboration, group decision making, and social business. Currently there are hundreds of tools designed to deal with the fact that, in order to succeed in our jobs, we are all much more dependent on one another, our fellow employees, customers, suppliers, and managers. Some of these tools are expensive, but others are available online for free (or with premium versions for a modest fee). Let's look more closely at some of these tools.

Email and Instant Messaging (IM)

Email and instant messaging (including text messaging) have been major communication and collaboration tools for interaction jobs. Their software operates

on computers, mobile phones, tablets, and other wireless devices and includes features for sharing files as well as transmitting messages. Many instant messaging systems allow users to engage in real-time conversations with multiple participants simultaneously. In recent years, email use has declined, with messaging and social media becoming preferred channels of communication.

Wikis

Wikis are a type of website that makes it easy for users to contribute and edit text content and graphics without any knowledge of web page development or programming techniques. The most well-known wiki is Wikipedia, the largest collaboratively edited reference project in the world. It relies on volunteers, makes no money, and accepts no advertising.

Wikis are very useful tools for storing and sharing corporate knowledge and insights. Enterprise software vendor SAP AG has a wiki that acts as a base of information for people outside the company, such as customers and software developers who build programs that interact with SAP software. In the past, those people asked and sometimes answered questions in an informal way on SAP online forums, but that was an inefficient system, with people asking and answering the same questions over and over.

Virtual Worlds

Virtual worlds, such as Second Life, are online 3D environments populated by “residents” who have built graphical representations of themselves known as avatars. Companies like IBM, Cisco, and Intel Corporation have used the online world for meetings, interviews, guest speaker events, and employee training. Real-world people represented by avatars meet, interact, and exchange ideas at these virtual locations using gestures, chat conversations, and voice communication.

Collaboration and Social Business Platforms

There are now software suites providing multifunction platforms for collaboration and social business among teams of employees who work together from different locations. The most widely used are Internet-based audio conferencing and videoconferencing systems, cloud collaboration services such as Google’s online services and tools, corporate collaboration systems such as Microsoft SharePoint and IBM Notes, and enterprise social networking tools such as Salesforce Chatter, Microsoft Yammer, Facebook Workplace, and IBM Connections. Some of these are now being used by educational institutions to deal with the impact of COVID-19 (see the Interactive Session on Technology).

Virtual Meeting Systems In an effort to reduce travel expenses and enable people in different locations to meet and collaborate, many companies, both large and small, are adopting videoconferencing and web conferencing technologies. More companies are using these systems to enable their employees to work remotely as well as in the office.

A videoconference allows individuals at two or more locations to communicate simultaneously through two-way video and audio transmissions. High-end videoconferencing systems feature **telepresence** technology, an integrated audio and visual environment that allows a person to give the appearance of being present at a location other than his or her true physical location. Free or low-cost Internet-based systems such as Microsoft Teams, Google Meet, and Amazon Chime are lower quality but improving, and they are very useful for smaller companies. Apple’s FaceTime is useful for one-to-one or small group videoconferencing. Zoom has emerged as a powerful videoconferencing tool with many features of high-end systems but is much less expensive and easier to use. Some of these tools are available on mobile devices.

INTERACTIVE SESSION TECHNOLOGY**Higher Education Redefined: Virtual Education at Ahlia University**

Collaboration in educational institutions usually involves cooperation among students, faculties, sponsors, and universities. Unlike the corporate world, higher education institutions have generally fallen behind in adopting the latest technologies and have continued to rely primarily on face-to-face (F2F) collaboration. F2F learning predominates because of the widespread view that it helps students stay focused, connect with each other, follow a standard timetable, and improve their physical wellness and mental alertness. Faculties believe that it enables them to better understand and assess their students' qualities and shortcomings and to fulfill their roles as mentors. F2F learning enables students to share their perspectives and have their questions addressed with a level of personal interaction that digital learning cannot replicate.

However, the COVID-19 pandemic has forced universities to rethink their assumptions. Although many universities worldwide had already begun to implement virtual learning before the pandemic hit, they were at varying degrees of implementation when the lockdowns began, and many had to rush to catch up.

Ahlia University Bahrain, the country's first private university, has effectively moved the bulk of its operations and services online: teaching, academic advising, career development services, counseling services, digital library services, payment services, document management, internship programs, student activities, and student complaint services. The university prepared itself for the online transition by implementing a series of measures spread over a period of two to three months. This included addressing infrastructural requirements for online education: optimum Internet speed, network upgrades, and licensing of Microsoft and other open-source or affordable software packages. Short videos and manuals on how to use different technologies and e-learning platforms were prepared to help students, faculties, and administrative bodies. Online training and support were provided through a hotline, its official website, virtual training courses, as well as through WhatsApp, other social media, and YouTube videos. The university provided its members with access to Microsoft Office 365 and a library of e-books as well. Security policies were implemented to enhance the protection of information and privacy. Faculty underwent intensive training to strengthen the

educational process itself and ensure that there would be no disruption in the event of an emergency.

Microsoft SharePoint, a web-based cloud collaboration tool for information storage and management, had already been implemented by the university toward the end of 2019. SharePoint is used to share and maintain up-to-date information and to expand awareness of the university's plans, decisions, events, rules, and policies among students and faculty. SharePoint has built cohesion, harnessed collective knowledge, and driven organizational efficiency by sharing common resources. It has also promoted collaboration between the internal units of the university and external bodies.

To conduct classes online, Ahlia University chose Microsoft Teams, a collaborative tool that allows users to chat, make calls, share documents, and conduct meetings. The university's IT team linked time tables and class lists from the university's Student Information System database, which stores and tracks all student information, including grades and attendance records, to Microsoft Teams using application programming interfaces (APIs) to import relevant course details. Using Microsoft Teams has enabled the university to create a secure environment where access to Teams chats and groups is barred to outsiders. All classes are recorded so students can review and recall all that has been taught. Advice, career development, and counseling services for mental health issues have also been made available through Microsoft Teams chats and calls.

The course details and class list from the Student Information System have also been linked with Moodle, an open-source learning management system. This has contributed significantly to mass enrollment and secured authentication. Through Moodle, instructors use quizzes, chat messages, forums, workshops, uploaded course materials, assessments, and surveys with tracking and reminder options to improve their students' efficiency in studying and to ensure the timely completion of assignments and courses.

To conduct online webinars and conferences with industry specialists and other universities, Ahlia University has used Zoom in addition to Microsoft Teams. As of September 2020, the university has participated in 9 webinars, 12 local conferences, and 34 international debates and conferences involving universities and industry specialists from Asia, the United Kingdom, the United States, and the Middle

East. Before the adoption of platforms like Zoom and Microsoft Teams, the number of Ahlia University's international collaborations were limited to only around 11 per year. Moving discussions online has greatly reduced expenses due to travel while facilitating the exchange of knowledge and encouraging academic contributions. The success of these collaborations has

prompted Ahlia University to consider focusing on online platforms even after the pandemic is over.

Sources: Ahlia University-Bahrain, "COVID-19," www.ahlia.edu.bh, accessed December 17, 2020; Ahlia University, "WTUN Webinar Series—Online Learning and Teaching: Lessons Learned from COVID-19-Part 2," Youtube.com, July 21, 2020; Ahlia University, "WTUN Webinar Series—Online Learning and Teaching: Lessons Learned from COVID-19-Part 1," Youtube.com, July 21, 2020; Ahlia University, "How Virtual Learning Is Becoming a Key Ingredient in Delivering University Courses," Facebook.com, April 9, 2020.

CASE STUDY QUESTIONS

1. How does Microsoft Teams promote efficient collaboration in universities and colleges?
2. What management, organization, and technology-based issues did Ahlia University address before pivoting its operations to deal with the pandemic?
3. What advantages has using collaboration tools provided for Ahlia University?

Case contributed by Subhashini Bhaskaran, Ahlia University

Cloud Collaboration Services Google offers many online tools and services, and some are suitable for collaboration. They include Google Drive, Google Docs, G Suite, and Google Sites. Most are free of charge.

Google Drive is a file storage and synchronization service for cloud storage, file sharing, and collaborative editing. Such web-based online file-sharing services allow users to upload files to secure online storage sites from which the files can be shared with others. Microsoft OneDrive and Dropbox are other leading cloud storage services. They feature both free and paid services, depending on the amount of storage space and administration required. Users are able to synchronize their files stored online with their local PCs and other kinds of devices, with options for making the files private or public and for sharing them with designated contacts.

Google Drive and Microsoft OneDrive are integrated with tools for document creation and sharing. OneDrive provides online storage for Microsoft Office documents and other files and works with Microsoft Office apps, both installed and on the web. It can share to Facebook as well. Google Drive is integrated with Google Docs, Sheets, and Slides (often called Google Docs), a suite of productivity applications that offer collaborative editing on documents, spreadsheets, and presentations. Google's cloud-based productivity suite for businesses, called G Suite, also works with Google Drive. Google Sites allows users to quickly create online team-oriented sites where multiple people can collaborate and share files.

Microsoft SharePoint and IBM Notes Microsoft SharePoint is a browser-based collaboration and document management platform, combined with a powerful search engine, that is installed on corporate servers. SharePoint has a web-based interface and close integration with productivity tools such as Microsoft Office. SharePoint software makes it possible for employees to share their documents and collaborate on projects using Office documents as the foundation.

SharePoint can be used to host internal websites that organize and store information in one central workspace to enable teams to coordinate work activities, collaborate on and publish documents, maintain task lists, implement workflows, and share information via wikis and blogs. Users are able to control versions of documents and document security. Because SharePoint stores and organizes information in one place, users can find relevant information

TABLE 2.4 ENTERPRISE SOCIAL NETWORKING SOFTWARE CAPABILITIES

SOCIAL SOFTWARE CAPABILITY	DESCRIPTION
Profiles	Ability to set up member profiles describing who individuals are, educational background, interests. Includes work-related associations and expertise (skills, projects, teams).
Content sharing	Share, store, and manage content including documents, presentations, images, and videos.
Feeds and notifications	Real-time information streams, status updates, and announcements from designated individuals and groups.
Groups and team workspaces	Establish groups to share information, collaborate on documents, and work on projects with the ability to set up private and public groups and to archive conversations to preserve team knowledge.
Tagging and social bookmarking	Indicate preferences for specific pieces of content, similar to the Facebook Like button. Tagging lets people add keywords to identify content they like.
Permissions and privacy	Ability to make sure private information stays within the right circles, as determined by the nature of relationships. In enterprise social networks, there is a need to establish who in the company has permission to see what information.

quickly and efficiently while working closely together on tasks, projects, and documents. Enterprise search tools help locate people, expertise, and content. SharePoint now features social tools.

IBM Notes (formerly Lotus Notes) is a collaborative software system with capabilities for sharing calendars, email, messaging, collective writing and editing, shared database access, and online meetings. Notes software installed on desktop or laptop computers obtains applications stored on an IBM Domino server. Notes is web-enabled and offers an application development environment so that users can build custom applications to suit their unique needs. Notes has also added capabilities for blogs, microblogs, wikis, online content aggregators, help desk systems, voice and video conferencing, and online meetings. IBM Notes promises high levels of security and reliability and the ability to retain control over sensitive corporate information.

Enterprise Social Networking Tools The tools we have just described include capabilities for supporting social business, but there are also more specialized social tools for this purpose, such as Salesforce Chatter, Microsoft Yammer, Facebook Workplace, and IBM Connections. Enterprise social networking tools create business value by connecting the members of an organization through profiles, updates, and notifications similar to Facebook features but tailored to internal corporate uses. Table 2.4 provides more detail about these internal social capabilities.

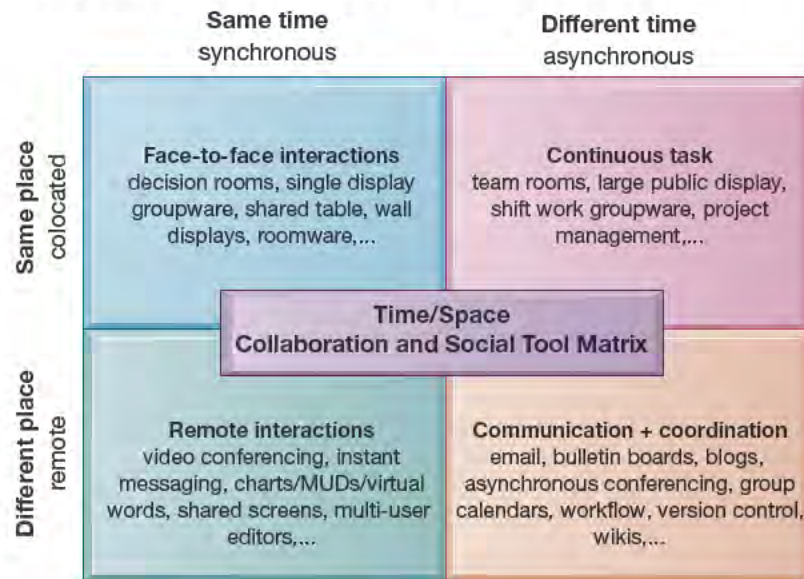
Although companies have benefited from enterprise social networking, internal social networking has not always been easy to implement. The chapter-ending case study addresses this topic.

Checklist for Managers: Evaluating and Selecting Collaboration and Social Software Tools

With so many collaboration and social business tools and services available, how do you choose the right collaboration technology for your firm? To answer this question, you need a framework for understanding just what problems these tools are designed to solve. One framework that has been helpful for us to talk about collaboration tools is the time/space collaboration and social tool matrix developed in the early 1990s by a number of collaborative work scholars (Figure 2.8).

FIGURE 2.8 THE TIME/SPACE COLLABORATION AND SOCIAL TOOL MATRIX

Collaboration and social technologies can be classified in terms of whether they support interactions at the same or different times or places, and whether these interactions are remote or colocated.



The time/space matrix focuses on two dimensions of the collaboration problem: time and space. For instance, you need to collaborate with people in different time zones, and you cannot all meet at the same time. Midnight in New York is noon in Mumbai, so this makes it difficult to have a videoconference (the people in New York are too tired). Time is clearly an obstacle to collaboration on a global scale.

Place (location) also inhibits collaboration in large global or even national and regional firms. Assembling people for a physical meeting is made difficult by the physical dispersion of distributed firms (firms with more than one location), the cost of travel, and the time limitations of managers.

The collaboration and social technologies we have just described are ways of overcoming the limitations of time and space. Using this time/space framework will help you to choose the most appropriate collaboration and teamwork tools for your firm. Note that some tools are applicable in more than one time/place scenario. For example, Internet collaboration suites such as IBM Notes have capabilities for both synchronous (instant messaging, meeting tools) and asynchronous (email, wikis, document editing) interactions.

Here's a "to-do" list to get started. If you follow these six steps, you should be led to investing in the correct collaboration software for your firm at a price you can afford and within your risk tolerance.

1. What are the collaboration challenges facing the firm in terms of time and space? Locate your firm in the time/space matrix. Your firm can occupy more than one cell in the matrix. Different collaboration tools will be needed for each situation.
2. Within each cell of the matrix where your firm faces challenges, exactly what kinds of solutions are available? Make a list of vendor products.

3. Analyze each of the products in terms of its cost and benefits to your firm. Be sure to include the costs of training in your cost estimates and the costs of involving the information systems department, if needed.
4. Identify the risks to security and vulnerability involved with each of the products. Is your firm willing to put proprietary information into the hands of external service providers over the Internet? Is your firm willing to expose its important operations to systems controlled by other firms? What are the financial risks facing your vendors? Will they be here in three to five years? What would be the cost of making a switch to another vendor in the event the vendor firm fails?
5. Seek the help of potential users to identify implementation and training issues. Some of these tools are easier to use than others.
6. Make your selection of candidate tools, and invite the vendors to make presentations.

2-4 What is the role of the information systems function in a business?

We've seen that businesses need information systems to operate today and that they use many different kinds of systems. But who is responsible for running these systems? Who is responsible for making sure the hardware, software, and other technologies used by these systems are running properly and are up to date? End users manage their systems from a business standpoint, but managing the technology requires a special information systems function.

The Information Systems Department

In all but the smallest of firms, the **information systems department** is the formal organizational unit responsible for information technology services. The information systems department is responsible for maintaining the hardware, software, data storage, and networks that comprise the firm's IT infrastructure. We describe IT infrastructure in detail in Chapter 5.

The information systems department consists of specialists, such as programmers, systems analysts, project leaders, and information systems managers. **Programmers** are highly trained technical specialists who write the software instructions for computers. **Systems analysts** constitute the principal liaisons between the information systems groups and the rest of the organization. It is the systems analyst's job to translate business problems and requirements into information requirements and systems. **Information systems managers** are leaders of teams of programmers and analysts, project managers, physical facility managers, telecommunications managers, or database specialists. They are also managers of computer operations and data entry staff. Also, external specialists, such as hardware vendors and manufacturers, software firms, and consultants, frequently participate in the day-to-day operations and long-term planning of information systems.

In many companies, the information systems department is headed by a **chief information officer (CIO)**. The CIO is a senior manager who oversees the use of information technology in the firm. Today's CIOs are expected to have a strong business background as well as information systems expertise and to play a leadership role in exploring new technologies and integrating technology into the firm's business strategy. Large firms today also have

positions for a chief security officer, chief knowledge officer, chief data officer, and chief privacy officer, all of whom work closely with the CIO.

The **chief security officer (CSO)** is in charge of information systems security for the firm and is responsible for enforcing the firm's information security policy (see Chapter 8). (Sometimes this position is called the chief information security officer [CISO] where information systems security is separated from physical security.) The CSO is responsible for educating and training users and information systems specialists about security, keeping management aware of security threats and breakdowns, and maintaining the tools and policies chosen to implement security.

Information systems security and the need to safeguard personal data have become so important that corporations collecting vast quantities of personal data have established positions for a **chief privacy officer (CPO)**. The CPO is responsible for ensuring that the company complies with existing data privacy laws.

The **chief knowledge officer (CKO)** is responsible for the firm's knowledge management program. The CKO helps design programs and systems to find new sources of knowledge or to make better use of existing knowledge in organizational and management processes.

The **chief data officer (CDO)** is responsible for enterprise-wide governance and utilization of information to maximize the value the organization can realize from its data. The CDO ensures that the firm is collecting the appropriate data to serve its needs, deploying appropriate technologies for analyzing the data, and using the results to support business decisions. This position arose to deal with the very large amounts of data organizations are now generating and collecting (see Chapter 6).

End users are representatives of departments outside the information systems group for whom applications are developed. These users are playing an increasingly large role in the design and development of information systems.

In the early years of computing, the information systems group was composed mostly of programmers who performed highly specialized but limited technical functions. Today, a growing proportion of staff members are systems analysts and network specialists, with the information systems department acting as a powerful change agent in the organization. The information systems department suggests new business strategies and new information-based products and services and coordinates both the development of the technology and the planned changes in the organization.

Information technology occupations in the United States were projected to grow 12 percent from 2018 through 2028, adding 546,200 new jobs (U.S. Bureau of Labor Statistics, 2020). Although all IT/IS occupations show above-average growth, jobs for information security analysts, data scientists, network analysts, machine learning engineers, cloud engineers, application developers, web developers, and business intelligence analysts are especially in demand. Demand for computer programmers has been declining, in part because the process of creating computer programs is becoming increasingly efficient with the growth of online software services, cloud computing, and outsourcing of coding to low-wage countries.

Organizing the Information Systems Function

There are many types of business firms, and there are many ways in which the IT function is organized within the firm. A very small company will not have a formal information systems group. It might have one employee who

is responsible for keeping its networks and applications running, or it might use consultants for these services. Larger companies will have a separate information systems department, which may be organized along several different lines, depending on the nature and interests of the firm. Our Learning Track describes alternative ways of organizing the information systems function within the business.

The question of how the information systems department should be organized is part of the larger issue of IT governance. **IT governance** includes the strategy and policies for using information technology within an organization. It specifies the decision rights and framework for accountability to ensure that the use of information technology supports the organization's strategies and objectives. How much should the information systems function be centralized? What decisions must be made to ensure effective management and use of information technology, including the return on IT investments? Who should make these decisions? How will these decisions be made and monitored? Firms with superior IT governance will have clearly thought out the answers.



2-5 How will MIS help my career?

Here is how Chapter 2 and this book can help you find a job as a sales support specialist.

The Company

Comprehensive Supplemental Insurance is a leading provider of individual supplemental accident, disability, health, and life insurance products. It is headquartered in Amsterdam and has an open position for an entry-level sales support specialist. The company offers supplemental insurance to complement existing employer benefits programs, maintaining a field sales force and corporate staff of over 5,000 people worldwide. It is known for investing in its employees and their career development.

Position Description

This position will provide overall systems, administrative, and data management support to the national sales organization for the company's division that markets to small businesses. Job responsibilities include:

- Daily administration and support of the firm's Salesforce.com customer relationship management system, including managing user setup, profiles and roles, and validating data.
- Assisting with data management and providing system training and ongoing support to the field.
- Preparing routine weekly, monthly, and quarterly sales and key performance indicator reports for sales management.
- Preparing agent commission reports and creating new reports as requested.
- Supporting various projects related to agent licensing and agent compensation.

Job Requirements

- Strong Excel skills plus some knowledge of data management
- Strong customer service skills
- Strong analytical, critical thinking, and communication skills
- Ability to multitask in a fast-paced environment
- College degree or two years of equivalent experience

Interview Questions

1. What do you know about customer relationship management? Have you ever worked with Salesforce.com? If so, what have you used the system for?
2. What do you know about data management? Have you ever worked with data management software? If so, what exactly have you done with it?
3. Tell us what you can do with Excel. What kinds of problems have you used Excel to solve? Did you take courses in Excel?
4. Have you ever worked in customer service? What exactly did you do? What do you think is required to take on a successful client-oriented role for this company's agents and customers?
5. Can you give an example of a client service challenge you had to face? How did you approach this challenge?

Author Tips

1. Review the section of this chapter on enterprise applications, the Chapter 9 discussion of customer relationship management, and Chapter 6 on data management.
2. Use the web and the professional networking site LinkedIn to find out more about the company, its insurance products and services, and the way it operates. Think about what it needs to do to support its agents and its customers and why customer relationship management and data management are so important. You might inquire about your responsibilities for data management in this job position.
3. Learn what you can about Salesforce.com, especially how to set up user profiles and roles and how to validate data. Indicate you want to learn more about Salesforce and work with this tool.
4. Inquire how you would be using Excel; for example, calculating agent commissions. If you've never done that before, show some of the Excel work you have done (and perhaps bring examples with you to the interview). Show that you would be eager to learn what you don't know in Excel to fulfill your job assignments.

REVIEW SUMMARY

2-1 What are business processes? How are they related to information systems?

A business process is a logically related set of activities that defines how specific business tasks are performed, and it represents a unique way in which an organization coordinates work, information, and knowledge. Managers need to pay attention to business processes because they determine how well the organization can execute its business, and they may be a source of strategic advantage. There are business processes specific to each of the major business functions, but many business processes are cross-functional. Information systems automate parts of business processes, and they can help organizations redesign and streamline these processes.

2-2 How do systems serve the different management groups in a business, and how do systems that link the enterprise improve organizational performance?

Systems serving operational management are transaction processing systems (TPS), such as payroll or order processing, that track the flow of the daily routine transactions necessary to conduct business. Management information systems (MIS) produce reports serving middle management by condensing information from TPS, and these are not highly analytical. Decision-support systems (DSS) support management decisions that are unique and rapidly changing using advanced analytical models. All of these types of systems provide business intelligence that helps managers and enterprise employees make more-informed decisions. These systems for business intelligence serve multiple levels of management and include executive support systems (ESS) for senior management that provide data in the form of graphs, charts, and dashboards delivered via portals using many sources of internal and external information.

Enterprise applications are designed to coordinate multiple functions and business processes. Enterprise systems integrate the key internal business processes of a firm into a single software system to improve coordination and decision making. Supply chain management systems help the firm manage its relationship with suppliers to optimize the planning, sourcing, manufacturing, and delivery of products and services. Customer relationship management (CRM) systems coordinate the business processes surrounding the firm's customers. Knowledge management systems enable firms to optimize the creation, sharing, and distribution of knowledge. Intranets and extranets are private corporate networks based on Internet technology that assemble information from disparate systems. Extranets make portions of private corporate intranets available to outsiders.

2-3 Why are systems for collaboration and social business so important, and what technologies do they use?

Collaboration is working with others to achieve shared and explicit goals. Social business is the use of internal and external social networking platforms to engage employees, customers, and suppliers, and it can enhance collaborative work. Collaboration and social business have become increasingly important in business because of globalization, the decentralization of decision making, and growth in jobs where interaction is the primary value-adding activity. Collaboration and social business enhance innovation, productivity, quality, and customer service. Tools for collaboration and social business include email and instant messaging, wikis, virtual meeting systems, virtual worlds, cloud-based file-sharing services, corporate collaboration systems such as Microsoft SharePoint and IBM Notes, and enterprise social networking tools such as Chatter, Yammer, and IBM Connections.

2-4 What is the role of the information systems function in a business?

The information systems department is the formal organizational unit responsible for information technology services. It is responsible for maintaining the hardware, software, data storage, and networks that comprise the firm's IT infrastructure. The department consists of specialists, such as programmers, systems analysts, project leaders, and information systems managers, and is often headed by a CIO.

Key Terms

Business intelligence, 77
Chief data officer (CDO), 97
Chief information officer (CIO), 96
Chief knowledge officer (CKO), 97
Chief privacy officer (CPO), 97
Chief security officer (CSO), 97
Collaboration, 86
Customer relationship management (CRM) systems, 83
Decision-support systems (DSS), 78
Digital dashboard, 80
Electronic business (e-business), 84
Electronic commerce (e-commerce), 84
E-government, 84
End users, 97
Enterprise applications, 81
Enterprise systems, 82
Executive support systems (ESS), 80
Information systems department, 96
Information systems managers, 96
Interorganizational system, 83
IT governance, 98
Knowledge management systems (KMS), 83
Management information systems (MIS), 78
Portal, 80
Programmers, 96
Social business, 88
Supply chain management (SCM) systems, 83
Systems analysts, 96
Teams, 87
Telepresence, 91
Transaction processing systems (TPS), 76

MyLab MIS

To complete the problems marked with MyLab MIS, go to the EOC Discussion Questions in MyLab MIS.

Review Questions

2-1 What are business processes? How are they related to information systems?

- Describe with examples how business processes are tied to the specific functional area of an organization.
- List and describe the business processes involved in fulfilling a customer order.

2-2 How do systems serve the different management groups in a business, and how do systems that link the enterprise improve organizational performance?

- Describe the characteristics of transaction processing systems (TPS) and the roles they play in a business.
- Describe the characteristics of business intelligence and explain why it is important in terms of decision-making.
- Identify which layer of management is most likely to use MIS and why.
- Describe how business intelligence systems also address the decision-making needs of senior management.
- Explain how a digital dashboard works.
- Explain how and why an organization might want to implement enterprise applications. What might be involved?

- Explain how customer relationship management helps an organization manage their interactions with customers.

2-3 Why are systems for collaboration and social business so important, and what technologies do they use?

- Define collaboration and social business and explain why they have become so important in business today.
- List and describe the business benefits of collaboration and social business.
- Describe a supportive organizational culture and business processes for collaboration.
- List and describe the various types of collaboration and social business tools.

2-4 What is the role of the information systems function in a business?

- Describe how the information systems function supports a business.
- Compare the roles played by programmers, systems analysts, information systems managers, the chief information officer (CIO), the chief security officer (CSO), the chief data officer (CDO), and the chief knowledge officer (CKO).

Discussion Questions

2-5 MyLab MIS How could information systems be used to support the order fulfillment process illustrated in Figure 2.1? What are the most important pieces of information these systems should capture? Explain your answer.

2-6 MyLab MIS Identify the steps that are performed in the process of selecting and checking out a book from your college library and

the information that flows among these activities. Diagram the process. Are there any ways this process could be changed to improve the performance of your library or your school? Diagram the improved process.

2-7 MyLab MIS Use the time/space collaboration and social tool matrix to classify the collaboration and social technologies used by the Sharp Corporation.

Hands-On MIS Projects

The projects in this section give you hands-on experience analyzing opportunities to improve business processes with new information system applications, using a spreadsheet to improve decision making about suppliers, and using Internet software to plan efficient transportation routes. Visit MyLab MIS to access this chapter's Hands-On MIS Projects.

Management Decision Problems

2-8 Heli is a manager at MoiMoiPixel, a Finnish startup developing free, fast-paced casual games for mobile platforms. MoiMoiPixel has been a success, gathering over 10 million returning players within a very short time. Nevertheless, the company is bleeding due to associated costs of hosting the game for all these players and paying salaries to the current staff of 40. Heli realizes the company doesn't know the players at all—when they play, why they play, where they live, and so on. Further, there is no communication at all with the players, and simply meeting at the coffee machine and sending emails is not enough to communicate efficiently with the growing staff. Help Heli by providing three suggestions for improving revenue and communication using information systems.

2-9 PetFence is based in Birmingham, in the United Kingdom, and makes custom-made fences to protect cats and dogs in household gardens. The company takes orders via email, designs the fence together with the client, and builds it in the client's garden. Finally, an invoice is sent to the client. PetFence depends on having just the right amount of necessary parts—too few and the work is delayed; too many and the storage fills up quickly. As business transactions are agreed on over the phone or by email, they sometimes forget to charge clients and pay suppliers. Moreover, email requests from prospective clients are often lost in the email flood, leading to missed sales opportunities and credibility issues. In what ways could information systems help PetFence? What issues can information systems not help them with?

Improving Decision Making: Using a Spreadsheet to Select Suppliers

Software skills: Spreadsheet date functions, data filtering, DVERAGE function

Business skills: Analyzing supplier performance and pricing

2-10 In this exercise, you will learn how to use spreadsheet software to improve management decisions about selecting suppliers. You will filter transactional data on suppliers based on several different criteria to select the best suppliers for your company.

You run a company that manufactures aircraft components. You have many competitors who are trying to offer lower prices and better service to customers, and you are trying to determine whether you can benefit from better supply chain management. In MyLab MIS, you will find a spreadsheet file that contains a list of all of the items that your firm has ordered from its suppliers during the past three months. The fields in the spreadsheet file include vendor name, vendor identification number, purchaser's order number, item identification number and item description (for each item ordered from the vendor), cost per item, number of units of the item ordered (quantity), total cost of each order, vendor's accounts payable terms, order date, and actual arrival date for each order.

Prepare a recommendation of how you can use the data in this spreadsheet database to improve your decisions about selecting suppliers. Some criteria to consider for identifying preferred suppliers include the supplier's track record for on-time deliveries, suppliers offering the best accounts payable terms, and

suppliers offering lower pricing when the same item can be provided by multiple suppliers. Use your spreadsheet software to prepare reports to support your recommendations.

Achieving Operational Excellence: Using Internet Software to Plan Efficient Transportation Routes

Software skills: Internet-based software

Business skills: Transportation planning

2-11 In this exercise, you will use Google Maps to map out transportation routes for a business and select the most efficient route.

You have just started working as a dispatcher for Kräiz Land Transport, a new trucking and delivery service based in Luxembourg. Your first assignment is to plan a delivery of office equipment and furniture from Esch-sur-Alzette to Niederanven. To guide your trucker, you need to know the most efficient route between the two cities. Use Google Maps to find the route that is the shortest distance between the two cities. Use Google Maps again to find the route that takes the least time. Compare the results. Which route should Cross-Country use?

Collaboration and Teamwork Project

Identifying Management Decisions and Systems

2-12 With a team of three or four other students, find a description of a manager in a corporation in *The Financial Times*, *Forbes*, *Fortune*, the *The Economist*, or another business publication, or do your research on the web. Gather information about what the manager does and the role he or she plays in the company. Identify the organizational level and business function where this manager works. Make a list of the kinds of decisions this manager has to make and the kind of information the manager would need for those decisions. Suggest how information systems could supply this information. If possible, use Google Docs and Google Drive or Google Sites to brainstorm, organize, and develop a presentation of your findings for the class.

Is Social Business Good Business?

CASE STUDY

As companies become more dispersed in the global marketplace, businesses are turning increasingly to workplace collaboration technology, including tools for internal social networking. These tools can promote employee collaboration and knowledge sharing, and help employees make faster decisions, develop more innovative ideas for products and services, and become more engaged in their work and their companies.

Adoption of internal enterprise social networking is also being driven by the flood of email that employees typically receive each day and are increasingly unable to handle. Hundreds of email messages must be opened, read, answered, forwarded, or deleted. For example, Winnipeg, Manitoba-based Duha Group, which produces color paint samples and color systems for paint companies across the globe, was able to eliminate 125,000 excess emails per year by adopting Salesforce Chatter social collaboration tools. Managing Director Emeric Duha, who used to receive 50 emails each morning from Asia, Europe, and Australia, now has a Chatter feed of everything going on in the company.

Another driver of enterprise social networking is “app fatigue.” In order to collaborate, many employees have to log on to numerous apps, creating additional work. Contemporary enterprise social networking systems often integrate multiple capabilities in one place.

Recent studies have found that collaboration tools could be effective in boosting efficiency and productivity, while enabling users to make better business decisions. The products also expanded the potential for innovation. Not all companies, however, are successfully using them. Implementation and adoption of enterprise social networking depends not only on the capabilities of the technology but on the organization's culture and the compatibility of these tools with the firm's business processes. The technologies won't provide benefits if they are applied to flawed business processes and organizational behaviors. Digital collaboration tools such as Microsoft Teams, Chatter, Yammer, Zoom, and WebEx added to email, texting, and messaging may enmesh employees in too many interactions, leaving even less time for in-depth individual thinking and problem-solving.

When firms introduce new social media technology (as well as other technologies), a sizable number of employees resist the new tools, clinging to old ways of working, including email, because they are more familiar and comfortable. There are companies where employees have duplicated communication on both social media and email, increasing the time and cost of performing their jobs. BASF, the world's largest chemical producer with subsidiaries and joint ventures in more than 80 countries, prohibited some project teams from using email to encourage employees to use new social media tools.

Social business requires a change in thinking, including the ability to view the organization more democratically in a flatter and more horizontal way. A social business is much more open to everyone's ideas. A secretary, assembly line worker, or sales clerk might be the source of the next big idea. As a result, getting people to espouse social business tools requires more of a “pull” approach, one that engages workers and offers them a significantly better way to work. In most cases, they can't be forced to use social apps.

Enterprise capabilities for managing social networks and sharing digital content can help or hurt an organization. Social networks can provide rich and diverse sources of information that enhance organizational productivity, efficiency, and innovation, or they can be used to support preexisting groups of like-minded people that are reluctant to communicate and exchange knowledge with outsiders. Productivity and morale will fall if employees use internal social networks to criticize others or pursue personal agendas.

Social business applications modeled on consumer-facing platforms such as Facebook and Twitter will not necessarily work well in an organization or organizational department that has incompatible objectives. Will the firm use social business for operations, human resources, or innovation? The social media platform that will work best depends on its specific business purpose. Additionally employees who have actively used Facebook and Twitter in their personal lives are often hesitant to use similar social tools for work

purposes because they see social media primarily as an informal, personal means of self-expression and communication with friends and family. Most managers want employees to use internal social tools to communicate informally about work, but not to discuss personal matters. Employees accustomed to Facebook and Twitter may have trouble imagining how they could use social tools without getting personal.

This means that instead of focusing on the technology, businesses should first identify how social initiatives will actually improve work practices for employees and managers. They need a detailed understanding of social networks: how people are currently working, with whom they are working, what their needs are, and measures for overcoming employee biases and resistance.

A successful social business strategy requires leadership and behavioral changes. Just sponsoring a social project is not enough—managers need to demonstrate their commitment to a more open, transparent work style. Employees who are used to collaborating and doing business in more traditional ways need an incentive to use social software. Changing an organization to work in a different way requires enlisting those most engaged and interested in helping, and designing and building the right workplace environment for using social technologies.

Management needs to ensure that the internal and external social networking efforts of the company are providing genuine value to the business. Content on the networks needs to be relevant, up-to-date, and easy to access; users need to be able to connect to people who have the information they need and would otherwise be out of reach or difficult to reach. Social business tools should be appropriate for the tasks on hand and the organization's business processes, and users need to understand how and why to use them.

For example, NASA's Goddard Space Flight Center had to abandon a custom-built enterprise social network called Spacebook because no one knew how its social tools would help people do their jobs. Spacebook had been designed and developed without taking into consideration the organization's culture and politics. This is not an isolated phenomenon. Dimension Data found that one-fourth of the 900 enterprises it surveyed focused more on the successful implementation of collaboration technology, rather than how it's used and adopted.

Despite the challenges associated with launching an internal social network, there are companies using these networks successfully. One company that has made social business work is Standard Bank, Africa's largest financial services provider, which operates in 33 countries (including 19 in Africa). Standard Bank has embraced social business to keep up with the pace of twenty-first-century business. The bank is using Microsoft Yammer to help it become a more dynamic organization.

Use of Yammer at Standard Bank started to take off in 2013, when the bank staged an important conference for its executives around the world and was looking for a collaborative platform for communicating conference logistics and posting content such as PowerPoint presentations. Many agencies and consultants who worked for the bank used Yammer and liked the tool. Once conference participants saw how intuitive and useful Yammer was, they wanted to use it in their own operations. Usage exploded, and the Yammer social network grew to over 20,000 users just six months after Standard Bank adopted the Enterprise version. Belinda Carreira, Standard Bank's Executive Head of Interactive Marketing, is also reaching out to departments most likely to benefit from enterprise social networking.

Standard Bank has over 400 Yammer social groups. Many are organized around projects and problem-solving, such as finding credit card solutions that work well in African countries. Yammer has become a platform for listening, where employees can easily share their concerns and insights. Yammer is also used for internal education. Yammer enables trainers to present more visual and varied material than in the past, including videos from the Internet. In some locations, the Internet may be down for half the day, but Standard's employees are still able to access Yammer on their mobile phones.

Carreira notes that successful adoption and use of a social tool such as Yammer will hit roadblocks without proper planning and organizational buy-in. Many factors must be considered. Carreira recommends that Yammer implementors work closely with their organization's IT department, risk and compliance teams, human resources, communications department, and executive leadership across the organization. In addition to internal resources, Standard Bank drew on expertise provided by Yammer and Microsoft.

Northwards Housing, a nonprofit organization providing affordable housing services in Manchester, England, has an open organizational culture, which encourages two-way communication and information transparency. Northwards has 340 employees, who do everything from rent collection to scheduling repairs and cleaning maintenance. The organization wanted a way of exchanging information internally and with its customers that was easy to use and did not require much time for technical updates. Northwards introduced Yammer in 2012 and now has 85 percent of employees engaged with the network.

Steve Finegan, Northward's Head of Business Effectiveness and Communication, believes executive support was critical to the network's growth. The Northwards CEO regularly participates in discussions, posts links to news stories of interest, and publishes a blog. The organization's executive directors, who were initially skeptical about Yammer's benefits, now actively post content on the network and answer questions.

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CASE STUDY QUESTIONS

- 2-13** Identify the management, organization, and technology factors affecting adoption of internal corporate social networks.
- 2-14** Compare the experiences implementing internal social networks of the organizations described in this case. Why was Standard Bank successful? What role did management play in this process?
- 2-15** Should all companies implement internal enterprise social networks? Why or why not?

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